

Frequency Transform in Digital Image

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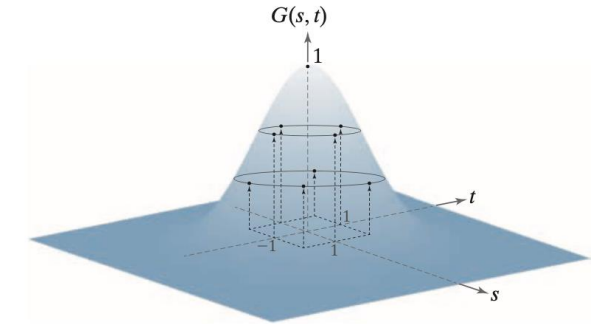
Previously...

Spatial filters

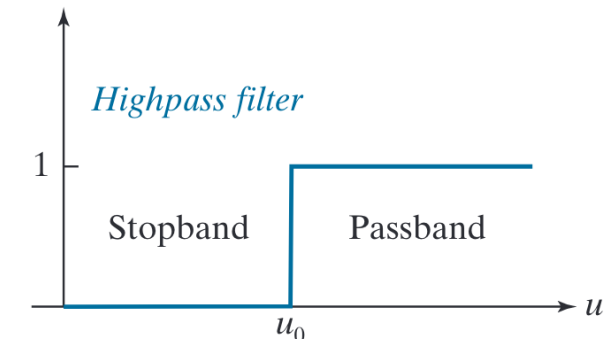
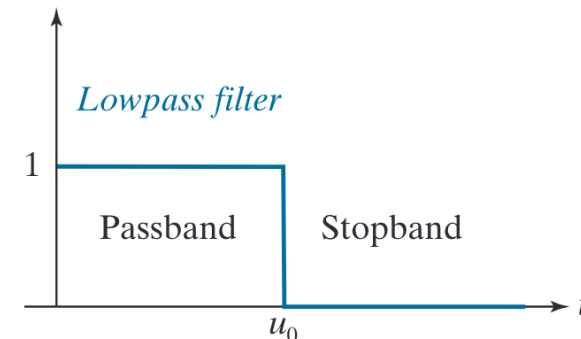
- Low pass filters
 - Box, Gaussian, Median, Max/Min
 - Denoising, smoothing
- High pass filters
 - Laplacian, Sobel, Gradient
 - Enhancement
- Bandpass/Bandreject filters

$$\frac{1}{4.8976} \times$$

0.3679	0.6065	0.3679
0.6065	1.0000	0.6065
0.3679	0.6065	0.3679



-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1



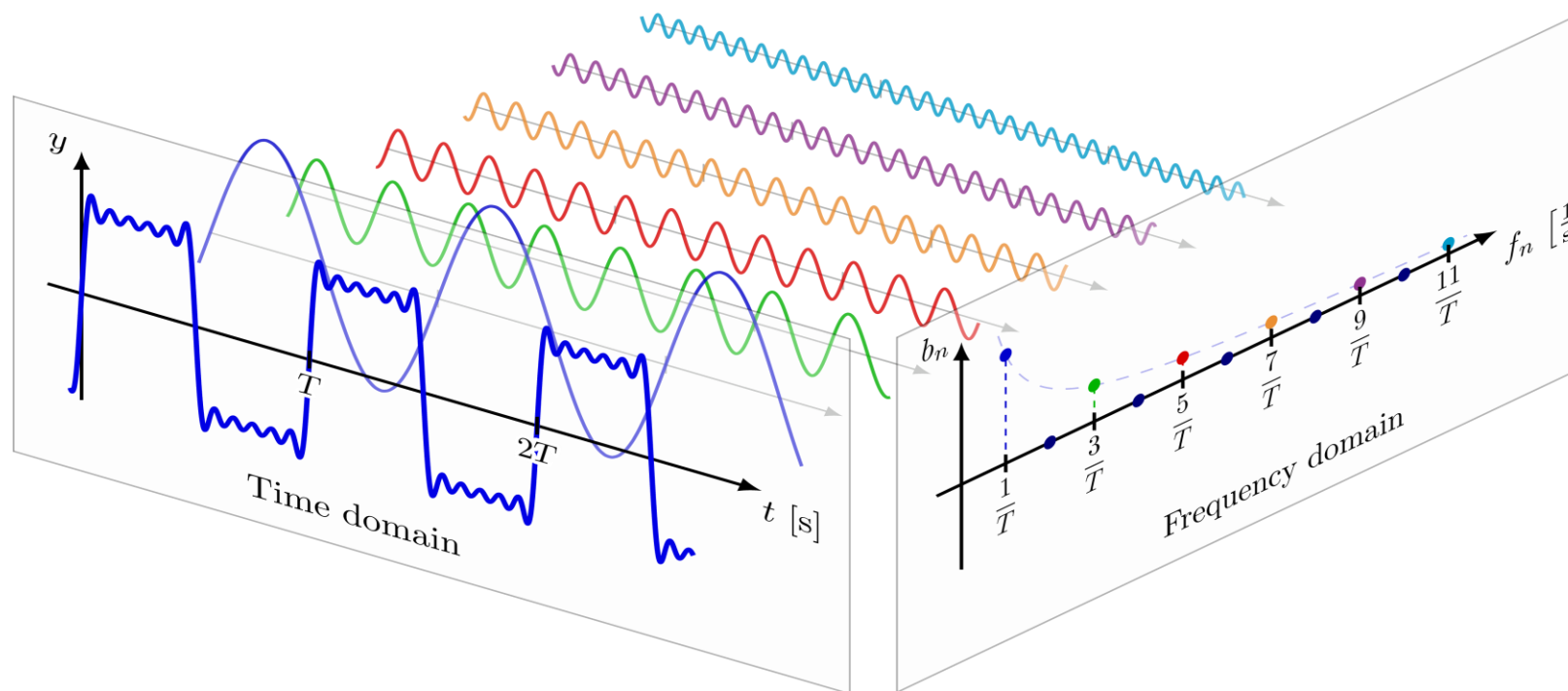
Frequency Domain Transformation

- Fourier Transform and its properties
- 2-D Fourier Transform
- 2-D Discrete Fourier Transform (2D-DFT)
- Fast Fourier Transform (FFT)
- Wavelet Transform

Background

– *Fourier series*

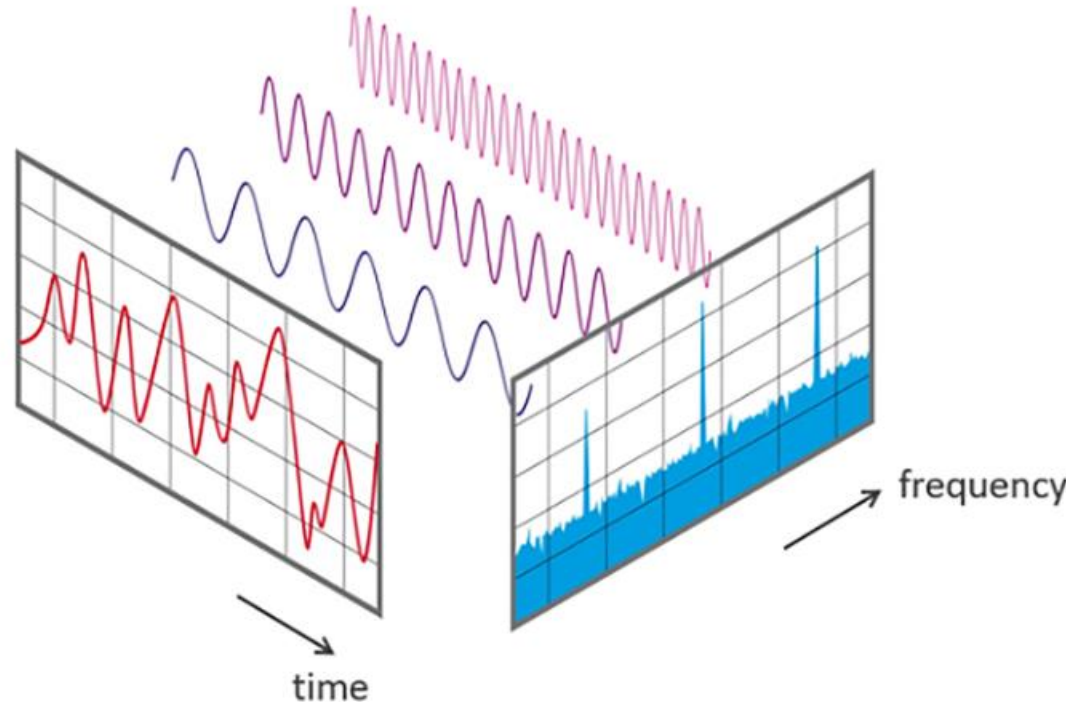
- Function that are periodic to sines and/or cosines of different frequencies
- Time domain **to** Frequency domain



Background

– *Fourier transform, FT*

Function that are
not periodic



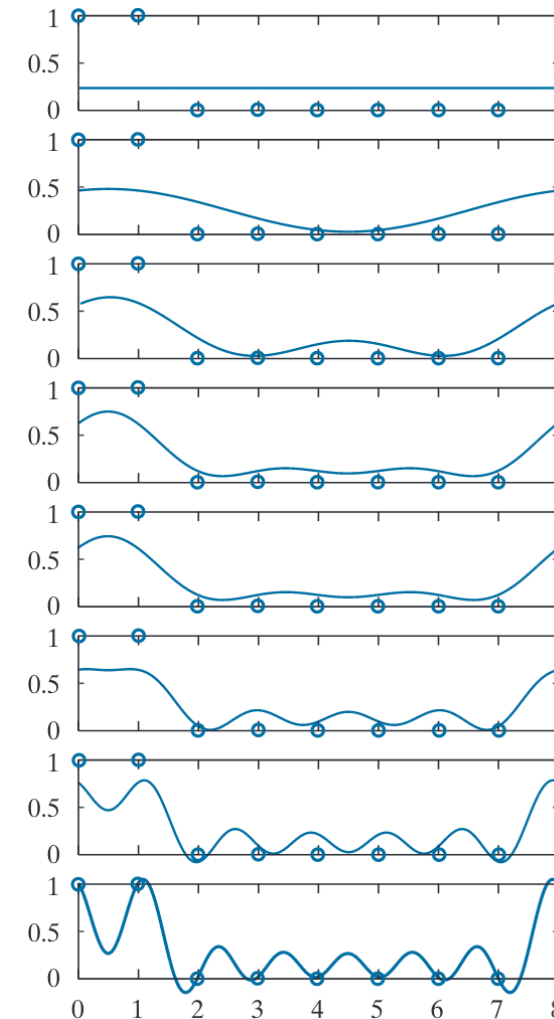
Continuous
function in
frequency domain

- Fourier's idea in 1807 was met with skepticism, but during the past century entire industries and academic disciplines have flourished as a result of it.
- It is used in designing electrical circuits, solving differential equations, signal processing, signal analysis, image processing & filtering.

Fourier Series

$$f = [1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0]^T$$

- Top: the average or DC value of f (for $u = 0$)
- Bottom: converging to f (for $u = 0, 1, \dots, 7$)



$$\hat{f}_0 = \frac{1}{4}$$

$$\hat{f}_1 = \hat{f}_0 + \frac{2 + \sqrt{2}}{16} \cos\left(\frac{\pi x}{4}\right) + \frac{\sqrt{2}}{16} \sin\left(\frac{\pi x}{4}\right)$$

$$\hat{f}_2 = \hat{f}_1 + \frac{1}{8} \cos\left(\frac{\pi x}{2}\right) + \frac{1}{8} \sin\left(\frac{\pi x}{2}\right)$$

...

FT of 1D Function

For a single variable, continuous function $f(x)$:

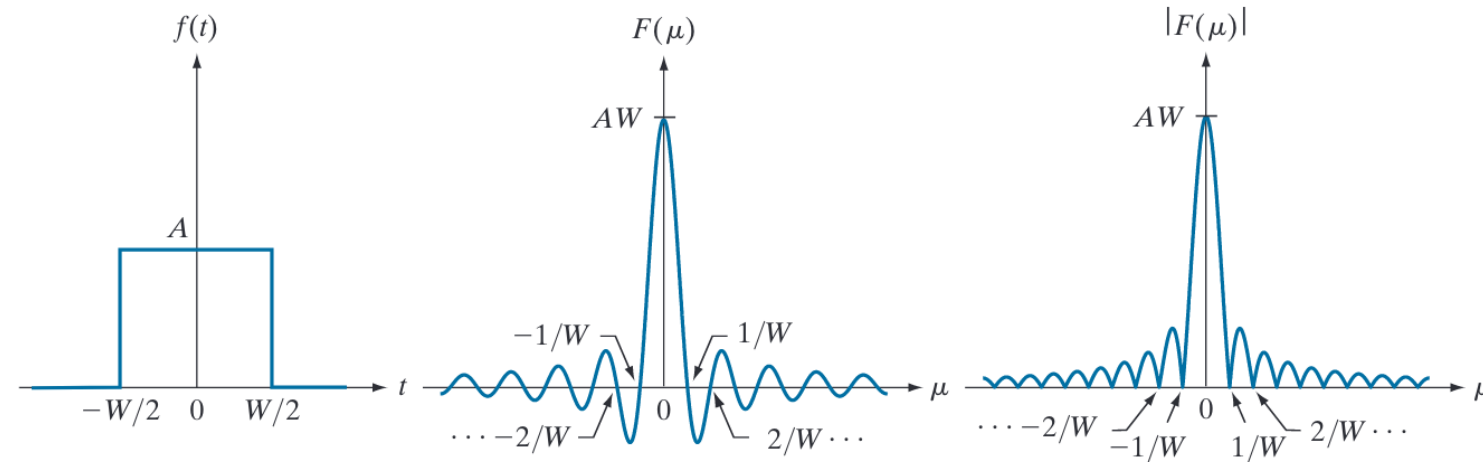
$$F(u) = \int_{-\infty}^{\infty} f(x) e^{-j2\pi ux} dx$$

u -space is called **frequency domain** because u determines the frequency

and the *inverse* Fourier transform is

$$f(x) = \int_{-\infty}^{\infty} F(u) e^{j2\pi ux} du$$

FT of the box function



$$\begin{aligned}
 F(\mu) &= \int_{-\infty}^{\infty} f(t) e^{-j2\pi\mu t} dt = \int_{-W/2}^{W/2} A e^{-j2\pi\mu t} dt \\
 &= \frac{-A}{j2\pi\mu} \left[e^{-j2\pi\mu t} \right]_{-W/2}^{W/2} = \frac{-A}{j2\pi\mu} \left[e^{-j\pi\mu W} - e^{j\pi\mu W} \right] \\
 &= \frac{A}{j2\pi\mu} \left[e^{j\pi\mu W} - e^{-j\pi\mu W} \right] \quad \boxed{e^{j\theta} = \cos\theta + j\sin\theta} \\
 &= AW \frac{\sin(\pi\mu W)}{(\pi\mu W)}
 \end{aligned}$$

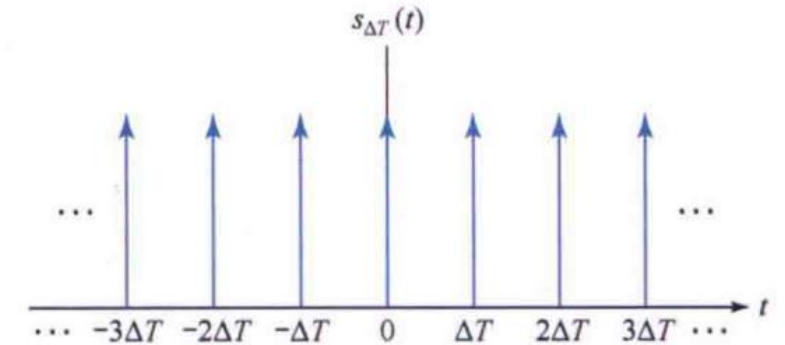
Rect (window) $\xleftrightarrow[\mathcal{F}^{-1}]{\mathcal{F}}$ sinc

FT of the impulse train

Impulse train function: $s_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} \delta(t - n\Delta T)$

Periodic function with period T can represent as

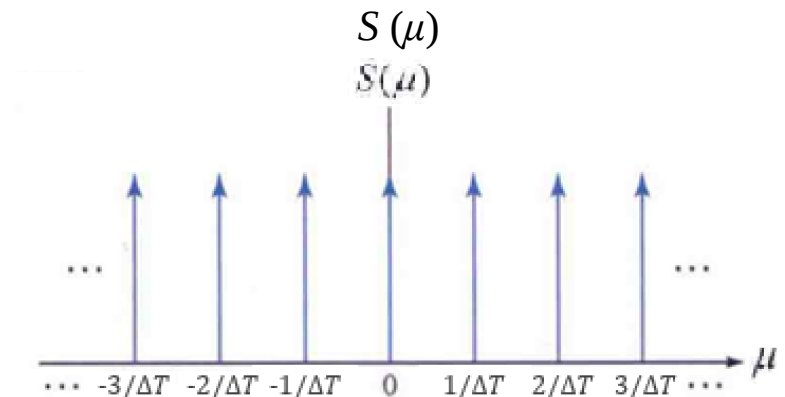
$$s_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} c_n e^{j\frac{2\pi n}{\Delta T}t} \text{ (Fourier series), where } c_n = \frac{1}{\Delta T}$$



$$\mathcal{F}\{e^{j\frac{2\pi n}{\Delta T}t}\} = \int_{-\infty}^{\infty} e^{j\frac{2\pi n}{\Delta T}t} e^{-j2\pi \mu t} dt = \delta(\mu - \frac{n}{\Delta T})$$

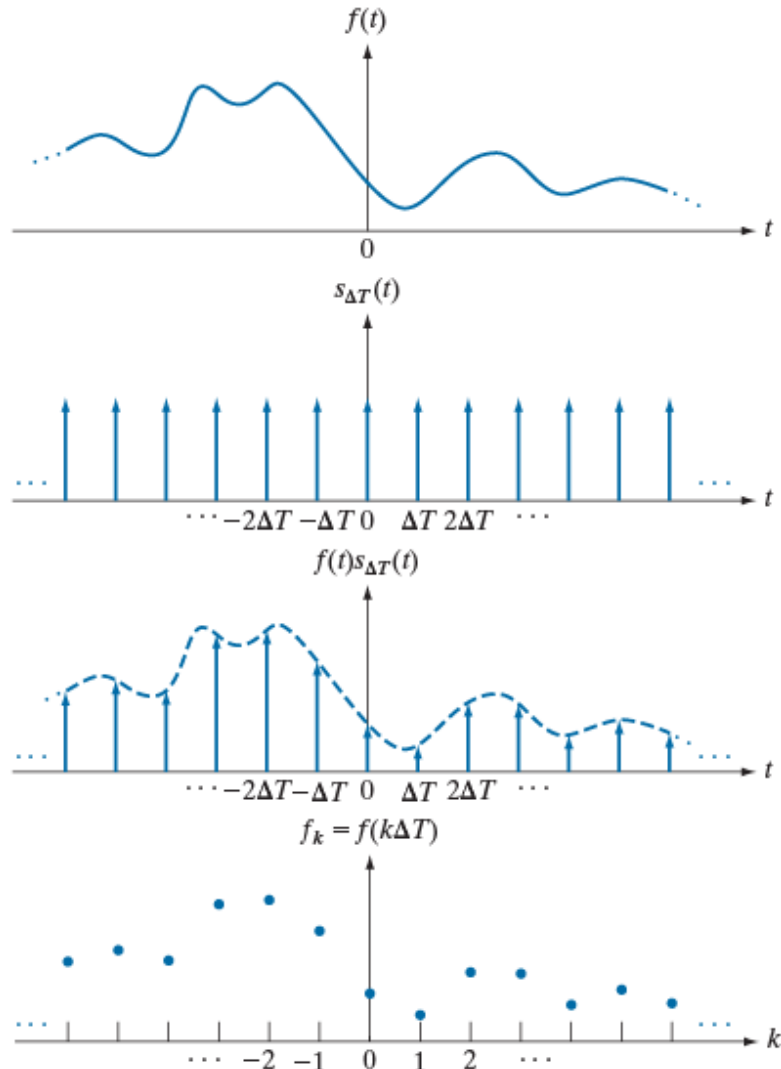
The Fourier transform of impulse train:

$$S(\mu) = \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} \delta(\mu - \frac{n}{\Delta T}).$$



impulse train(sample) $\longleftrightarrow$ copy and shift

FT of a discrete signal



$$\tilde{F}(\mu) = \mathcal{F}\{\tilde{f}(t)\} = \mathcal{F}\{f(t)s_{\Delta T}(t)\} = (F \star S)(\mu)$$

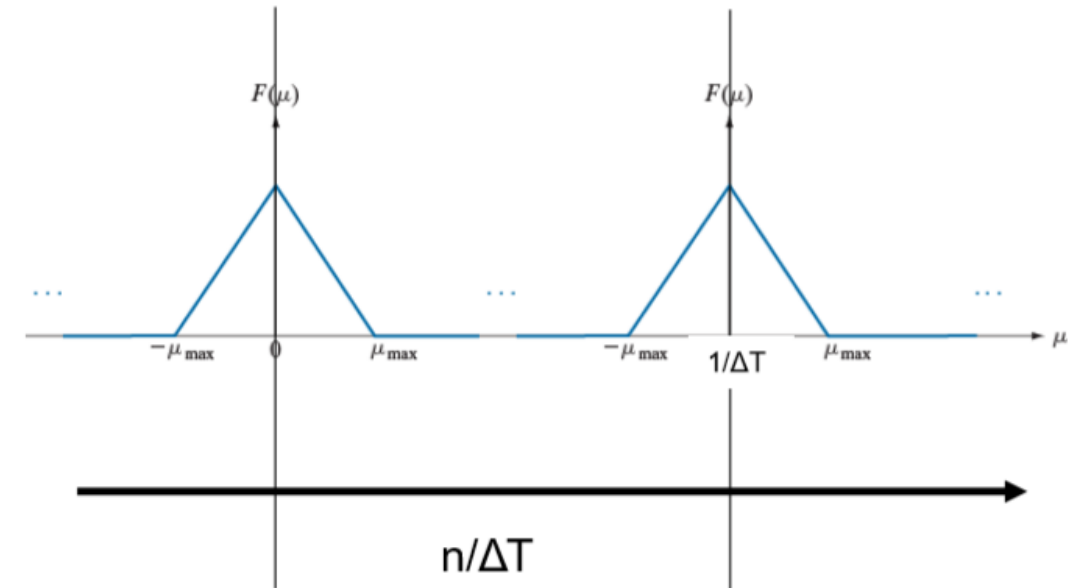
$$S(\mu) = \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} \delta\left(\mu - \frac{n}{\Delta T}\right)$$

$$\begin{aligned} \tilde{F}(\mu) &= (F \star S)(\mu) = \int_{-\infty}^{\infty} F(\tau) S(\mu - \tau) d\tau \\ &= \frac{1}{\Delta T} \int_{-\infty}^{\infty} F(\tau) \sum_{n=-\infty}^{\infty} \delta\left(\mu - \tau - \frac{n}{\Delta T}\right) d\tau \\ &= \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} F(\tau) \delta\left(\mu - \tau - \frac{n}{\Delta T}\right) d\tau \\ &= \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} F\left(\mu - \frac{n}{\Delta T}\right) \end{aligned}$$

The Sampling Theorem

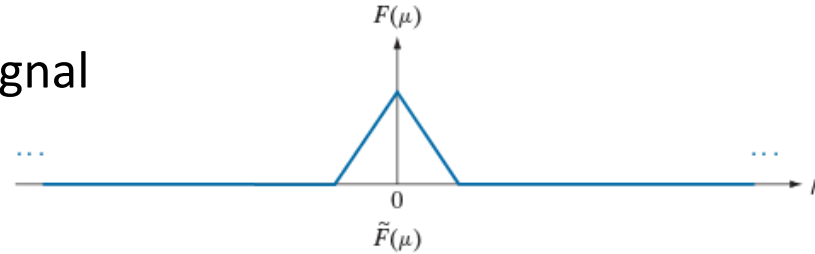
$$\tilde{F}(\mu) = \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} F\left(\mu - \frac{n}{\Delta T}\right)$$

- The spectrum of the discrete signal **consists of repetitions of the spectrum** of the continuous signal every $1/\Delta T$.
- Shorter sample interval means larger interval in frequency domain

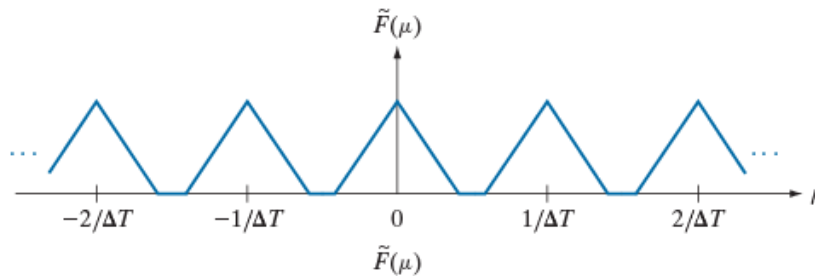


The Sampling Theorem

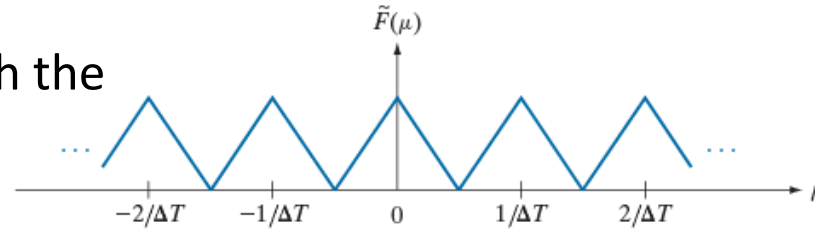
FT of a continuous signal



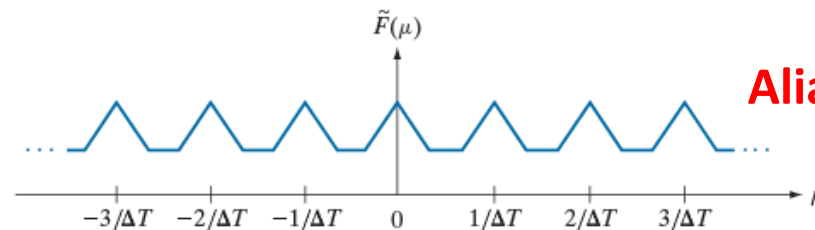
Over-sampling



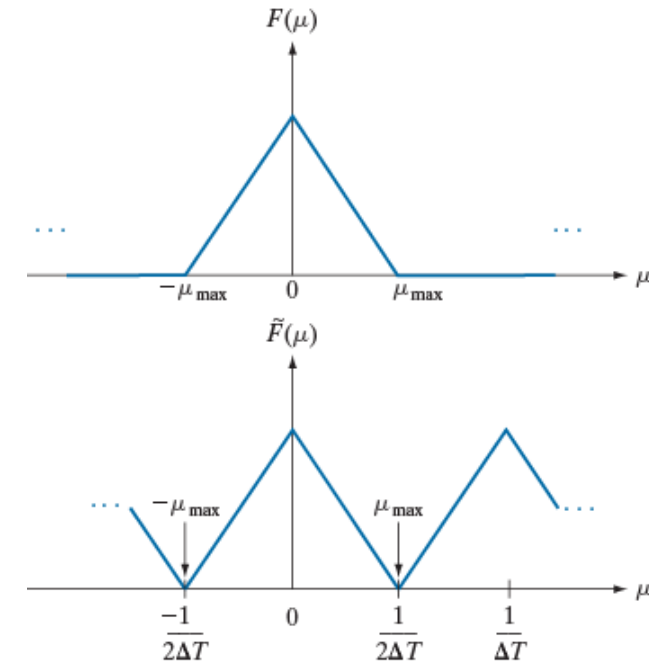
Critical sampling with the Nyquist frequency



Under-sampling



Aliasing!



The **Nyquist criterion** should be satisfied:

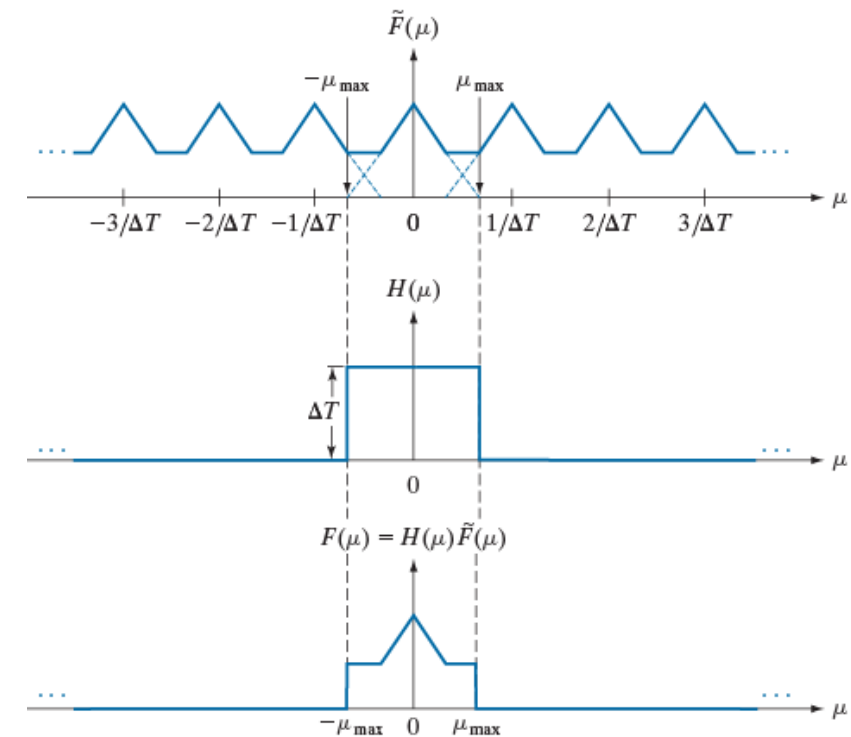
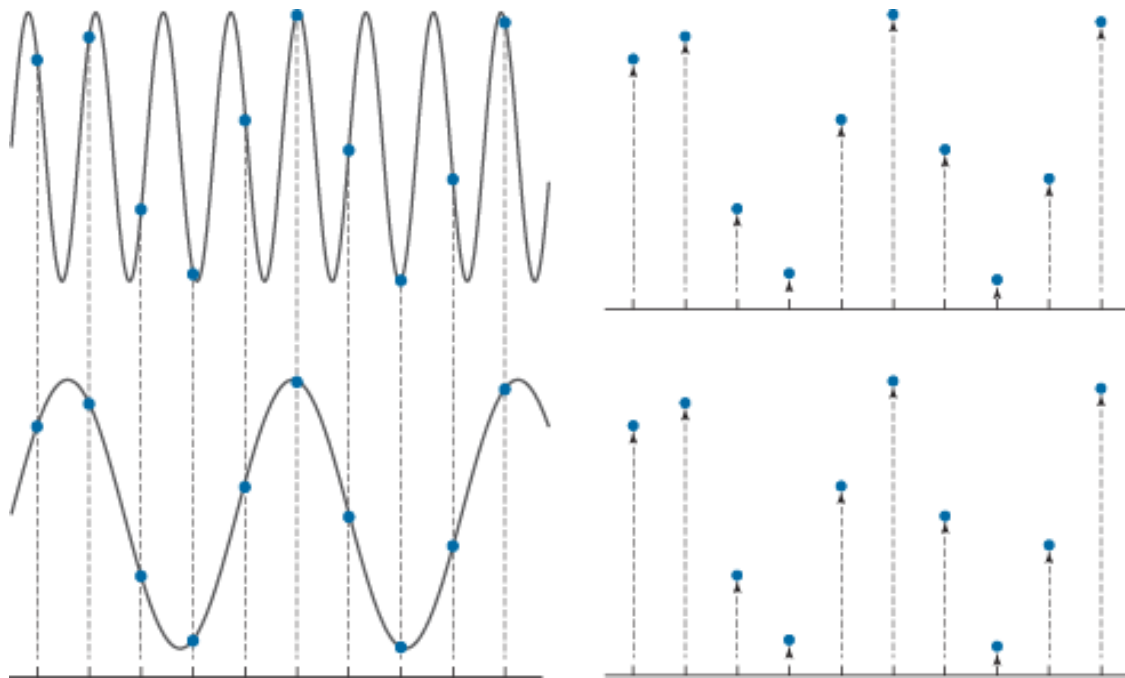
$$\frac{1}{\Delta T} \geq 2\mu_{\max}$$

The Sampling Theorem



Aliasing

- In frequency domain, use **rect function** to select the basic frequency component.
- Under aliasing, the **reconstruction** of the continuous signal **not correct**.



Discrete Fourier Transform (DFT)

In reality, the signal is **finite** and **discrete** → use Discrete Fourier Transform

For a M-length discrete signal :

$$F(u) = \frac{1}{M} \sum_{x=0}^{M-1} f(x) e^{-j2\pi ux/M} \quad u = 0, 1, 2, \dots, M-1.$$

↑
The multiplier sometimes is placed in front of the IDFT

↖
each of the M terms is called a *frequency component* of the transform

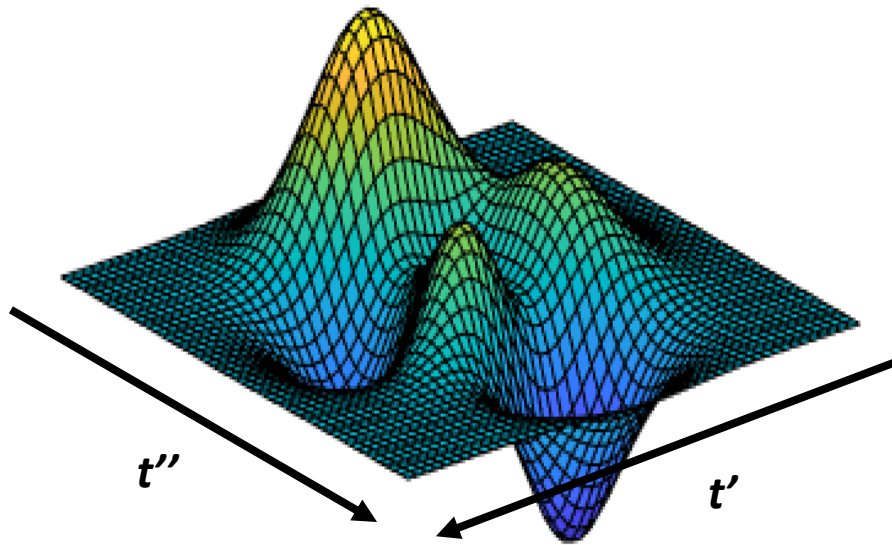
Similarly, given $F(u)$, we can obtain the original function back using the IDFT:

$$f(x) = \sum_{u=0}^{M-1} F(u) e^{j2\pi ux/M} \quad x = 0, 1, 2, \dots, M-1.$$

1D FT/DFT Summary

- Fourier Transform
 - *time domain to frequency domain*
- **Sampling and Aliasing**
 - *Nyquist criterion* $\frac{1}{\Delta T} \geq 2\mu_{\max}$
- **DFT** pairs between signal $f(x)$ and $F(u)$
 - One Variable : Discrete 1-D Signal
- *Image as a Discrete Function?*
 - **Extensions to 2D image, 2 variables FT, 2-D DFT**

The 2-D Fourier transform

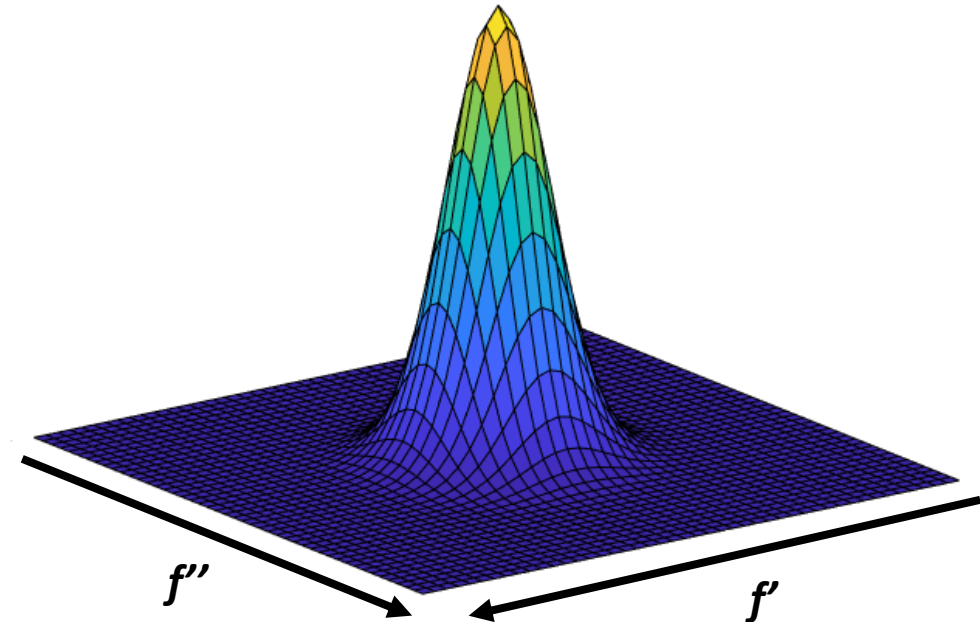
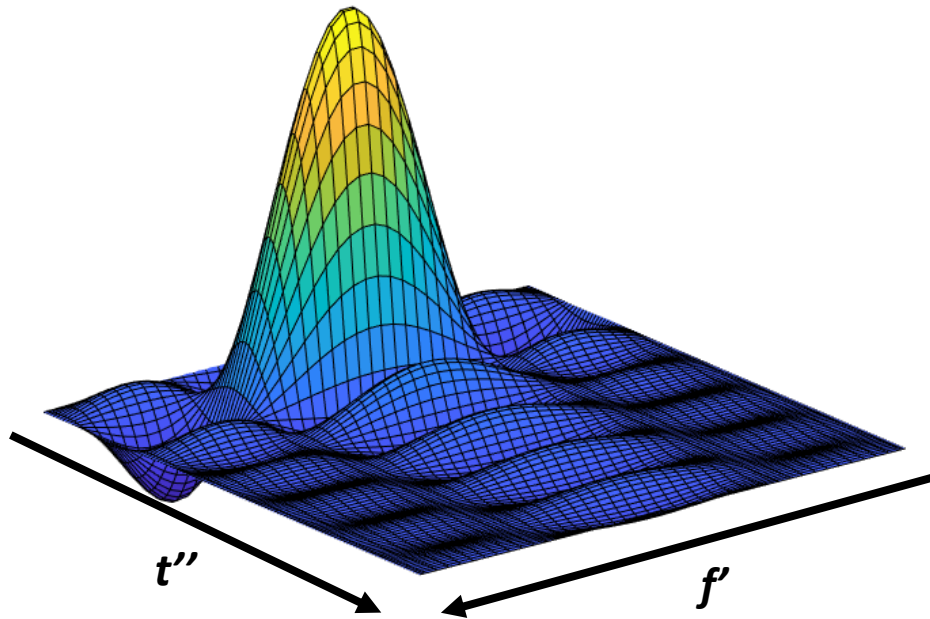


- Begin with a 2-D array of data: t' by t''
- Since this data is two dimensional, we say that the data is in the ***spatial domain***
- The process: first Fourier transform in one direction: t' , finally in the t'' direction

FT along two directions

First Fourier transform in one direction: t'

Then in the t'' direction



The spike corresponds to the intensity and location of the frequency within the 2-D frequency domain

FT of 2D Function

Extending to two variables, u and v :

$$F(u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) e^{-j2\pi(ux+vy)} dx dy$$

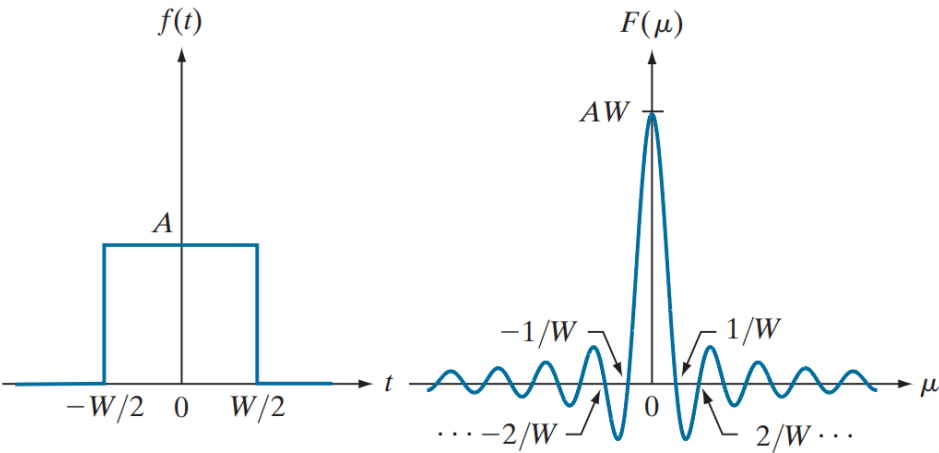
u and v determines the frequency

and, similarly for the inverse transform,

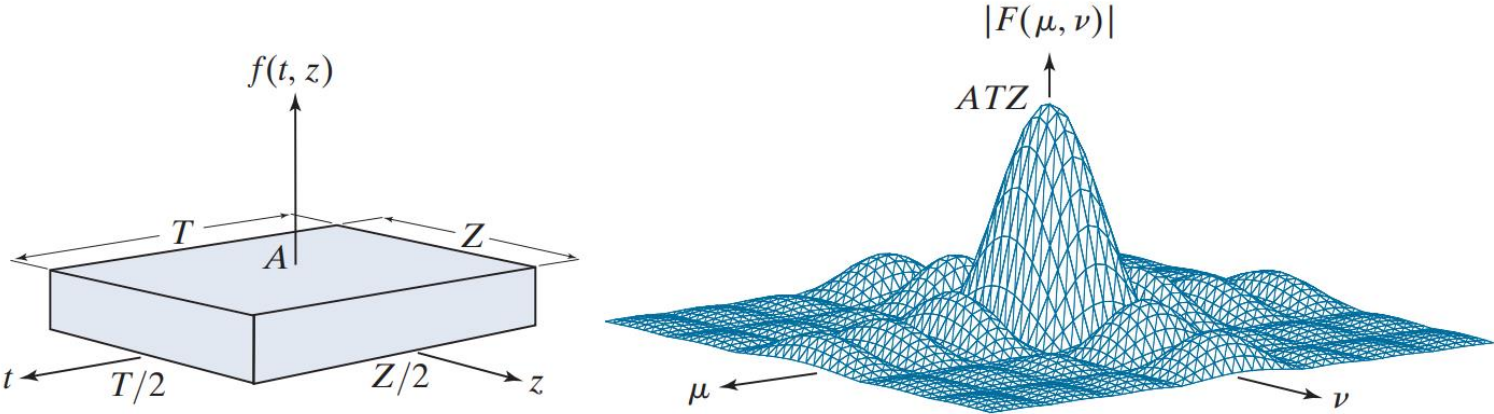
$$f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(u, v) e^{j2\pi(ux+vy)} du dv$$

Example of 2D FT

1D



2D



Frequency Domain in Medical Image

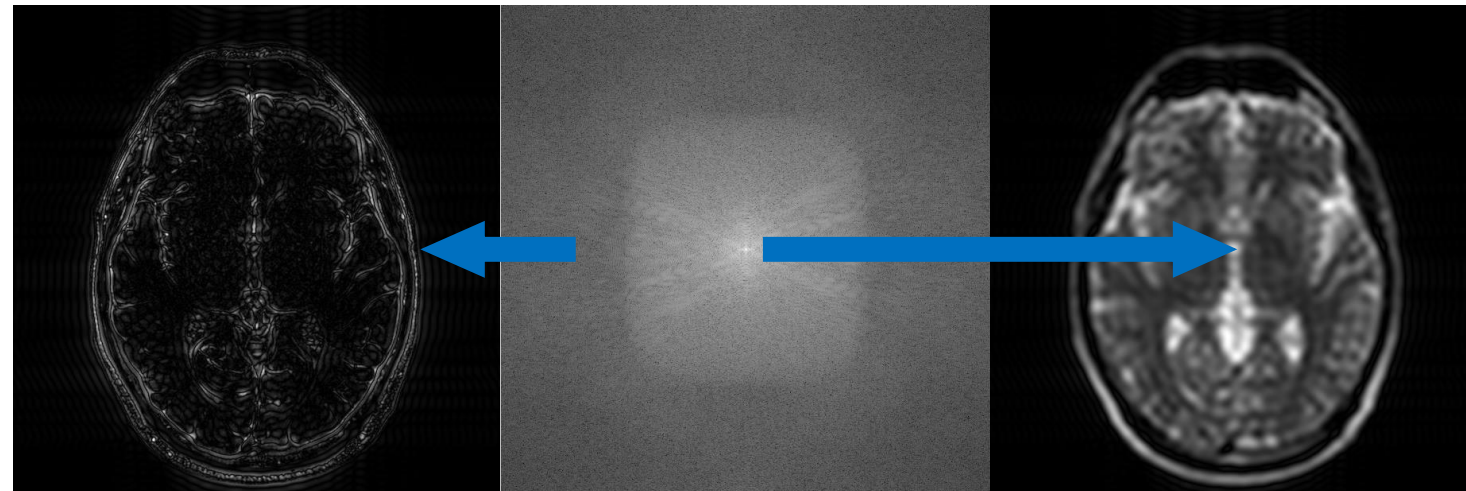
Spatial domain: x , distance (usually in pixels)

Frequency domain: can be measured with either:

- ω , **angular frequency** in radians per unit distance ,or

- f , **rotational frequency** in cycles per unit distance. $\omega = 2\pi f$

Example: K-space is the Fourier transform of the MR image



2-D Discrete Fourier Transform (2-D DFT)

The DFT of a function (image) $f(x, y)$ of size $M \times N$ is given by the equation

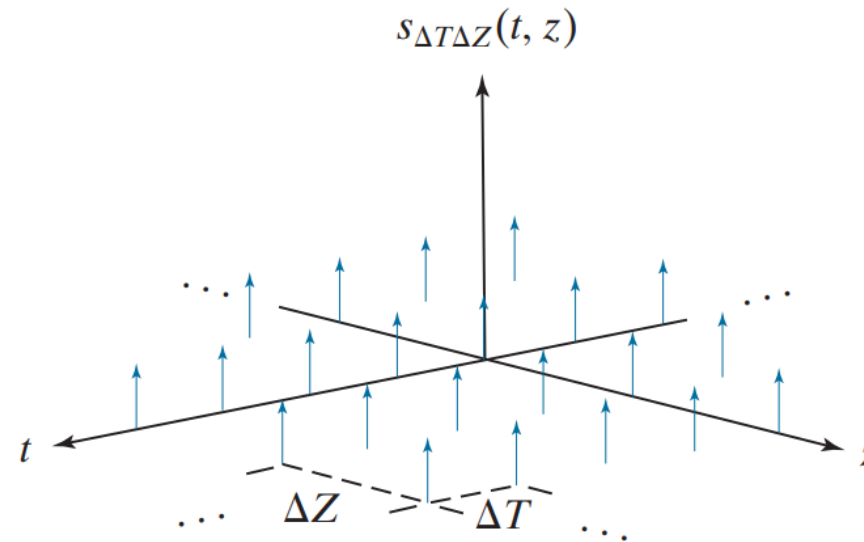
$$F(u, v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)} \quad \begin{aligned} u &= 0, 1, 2, \dots, M-1, \\ v &= 0, 1, 2, \dots, N-1. \end{aligned}$$

Similarly, the IDFT is

$$f(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)} \quad \begin{aligned} x &= 0, 1, 2, \dots, M-1, \\ y &= 0, 1, 2, \dots, N-1. \end{aligned}$$

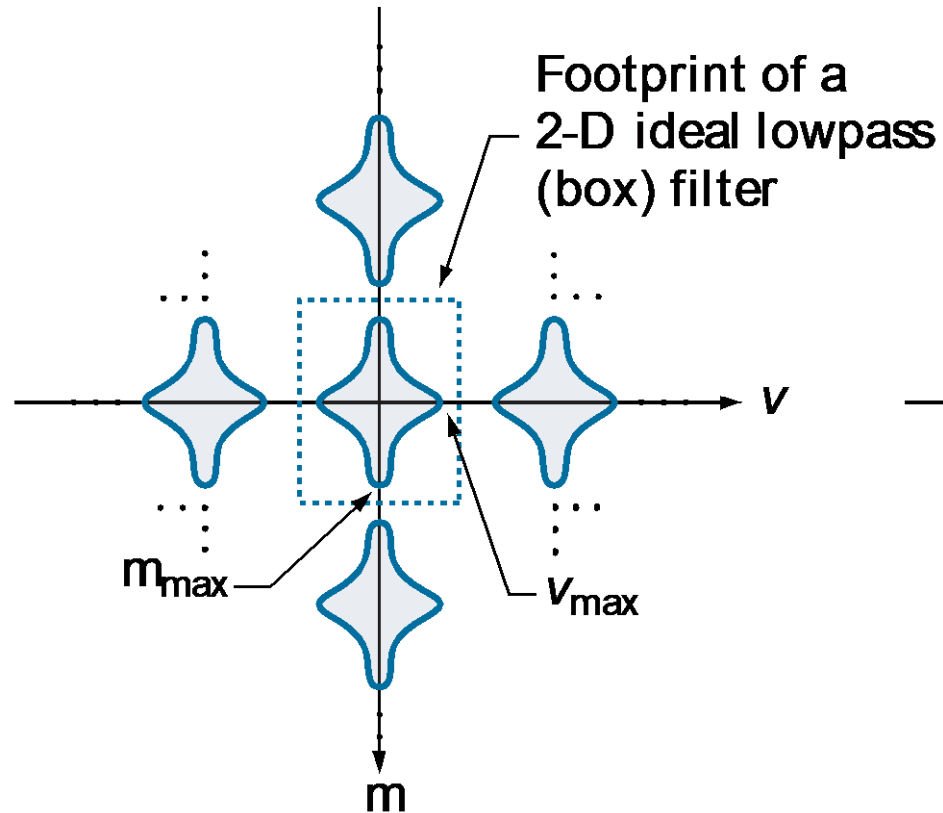
2-D Sampling Theorem

2-D impulse trains function

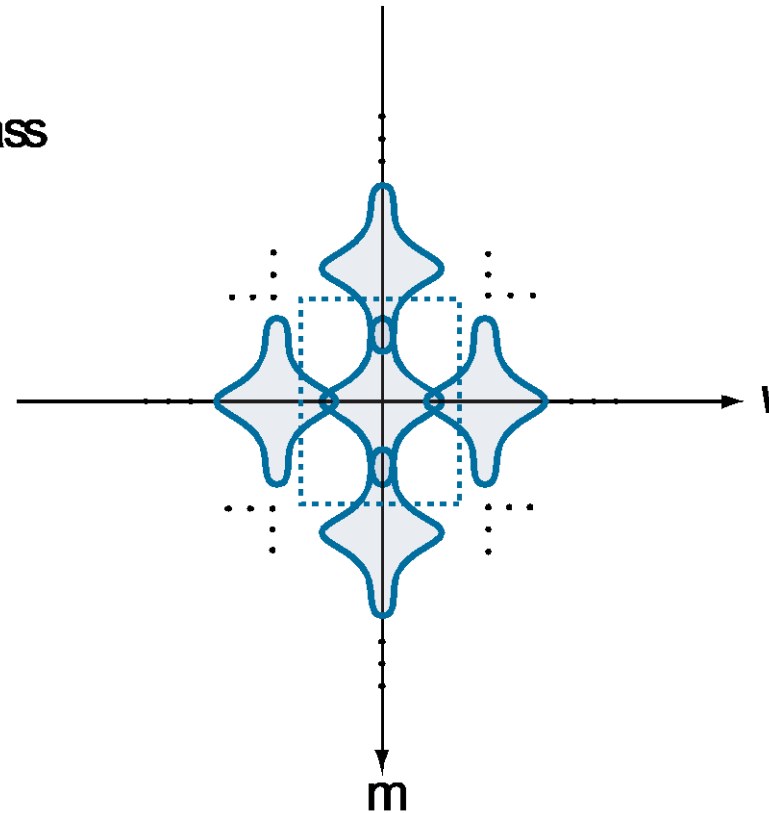


$$S_{DT\Delta Z}(t, z) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \delta(t - m\Delta T, z - n\Delta Z)$$

2-D Sampling Theorem



Over-sampled



Under-sampled

$$\frac{1}{\Delta T} \geq 2\mu_{\max}$$

$$\frac{1}{\Delta Z} \geq 2v_{\max}$$

2-D Sampling Theorem



Size 772x548

Negligible aliasing



Resize to 33%

Clearly aliasing

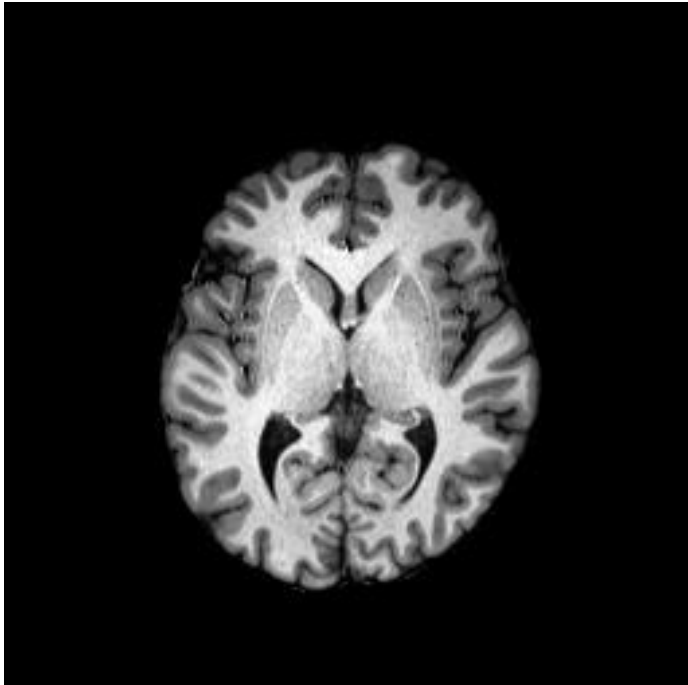


Average filter prior to resizing

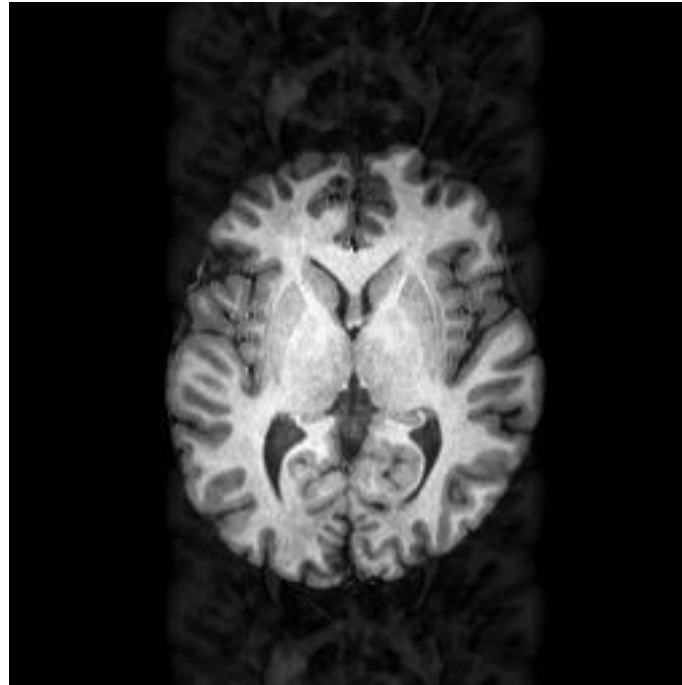
No objectionable aliasing

2-D Sampling Theorem

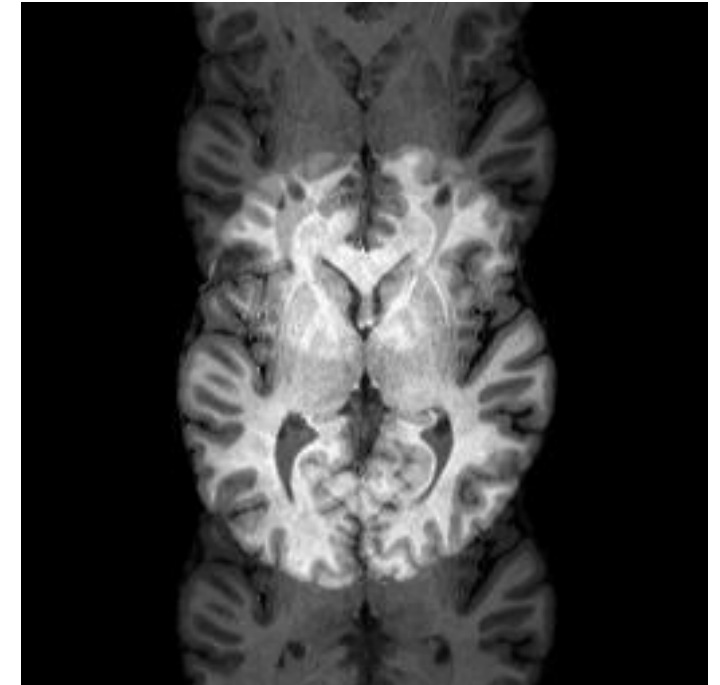
No aliasing



Low aliasing



Strong aliasing



Properties of the 2-D DFT

Name	Expression(s)
1) Discrete Fourier transform (DFT) of $f(x, y)$	$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$
2) Inverse discrete Fourier transform (IDFT) of $F(u, v)$	$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)}$
3) Spectrum	$ F(u, v) = [R^2(u, v) + I^2(u, v)]^{1/2} \quad R = \text{Real}(F); I = \text{Imag}(F)$
4) Phase angle	$\phi(u, v) = \tan^{-1} \left[\frac{I(u, v)}{R(u, v)} \right]$
5) Polar representation	$F(u, v) = F(u, v) e^{j\phi(u, v)}$
6) Power spectrum	$P(u, v) = F(u, v) ^2$
7) Average value	$\bar{f} = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) = \frac{1}{MN} F(0, 0)$
8) Periodicity (k_1 and k_2 are integers)	$F(u, v) = F(u + k_1 M, v) = F(u, v + k_2 N)$ $= F(u + k_1, v + k_2 N)$ $f(x, y) = f(x + k_1 M, y) = f(x, y + k_2 N)$ $= f(x + k_1, y + k_2 N)$

Name	Expression(s)
9) Convolution	$(f \star h)(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n) h(x - m, y - n)$
10) Correlation	$(f \star h)(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f^*(m, n) h(x + m, y + n)$
11) Separability	The 2-D DFT can be computed by computing 1-D DFT transforms along the rows (columns) of the image, followed by 1-D transforms along the columns (rows) of the result. See Section 4.11.
12) Obtaining the IDFT using a DFT algorithm	$MNf^*(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F^*(u, v) e^{-j2\pi(ux/M + vy/N)}$ This equation indicates that inputting $F^*(u, v)$ into an algorithm that computes the forward transform (right side of above equation) yields $MNf^*(x, y)$. Taking the complex conjugate and dividing by MN gives the desired inverse. See Section 4.11.

Properties of the 2-D DFT

- Zero-frequency term

The value of the transform at $(u, v) = (0, 0)$ is

$$F(0,0) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y)$$

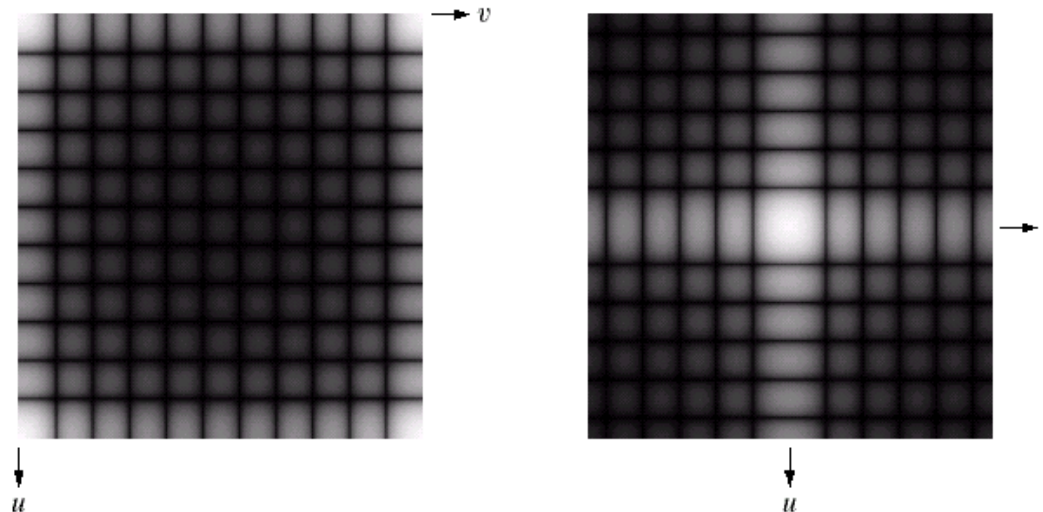
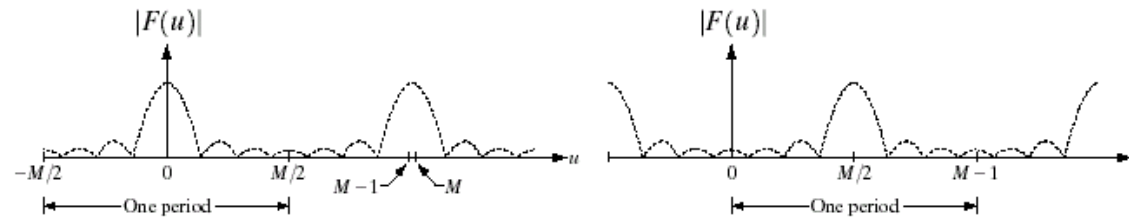
which is the average of $f(x, y)$. Sometimes $F(0, 0)$ is called the **dc** component of the spectrum.

Properties of the 2-D DFT

- Periodicity

$$F(u, v) = F(u + M, v) = F(u, v + N) = F(u + M, v + N)$$

$$f(x, y) = f(x + M, y) = f(x, y + N) = f(x + M, y + N)$$



Properties of the 2-D DFT

- Distributivity and scaling**

$$\mathfrak{F}[f_1(x, y) + f_2(x, y)] = \mathfrak{F}[f_1(x, y)] + \mathfrak{F}[f_2(x, y)]$$

and, in general

$$\mathfrak{F}[f_1(x, y) \cdot f_2(x, y)] \neq \mathfrak{F}[f_1(x, y)] \cdot \mathfrak{F}[f_2(x, y)]$$

For two scalars a and b ,

$$af(x, y) \Leftrightarrow aF(u, v)$$

$$f(ax, by) \Leftrightarrow \frac{1}{|ab|} F(u/a, v/b)$$

Properties of the 2-D DFT

- Shifting**

$$f(x, y)e^{j2\pi(u_0x/M+v_0y/N)} \Leftrightarrow F(u - u_0, v - v_0)$$

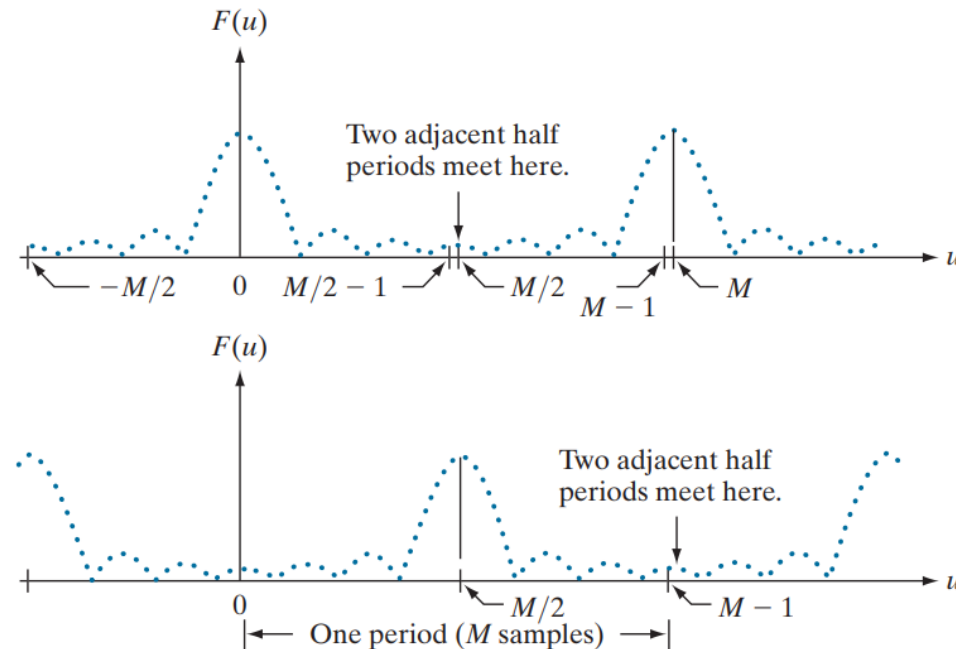
and

$$f(x - x_0, y - y_0) \Leftrightarrow F(u, v)e^{-j2\pi(ux_0/M+vy_0/N)}$$

$$f(x, y)e^{j2\pi u_0x/M} \Leftrightarrow F(u - u_0)$$

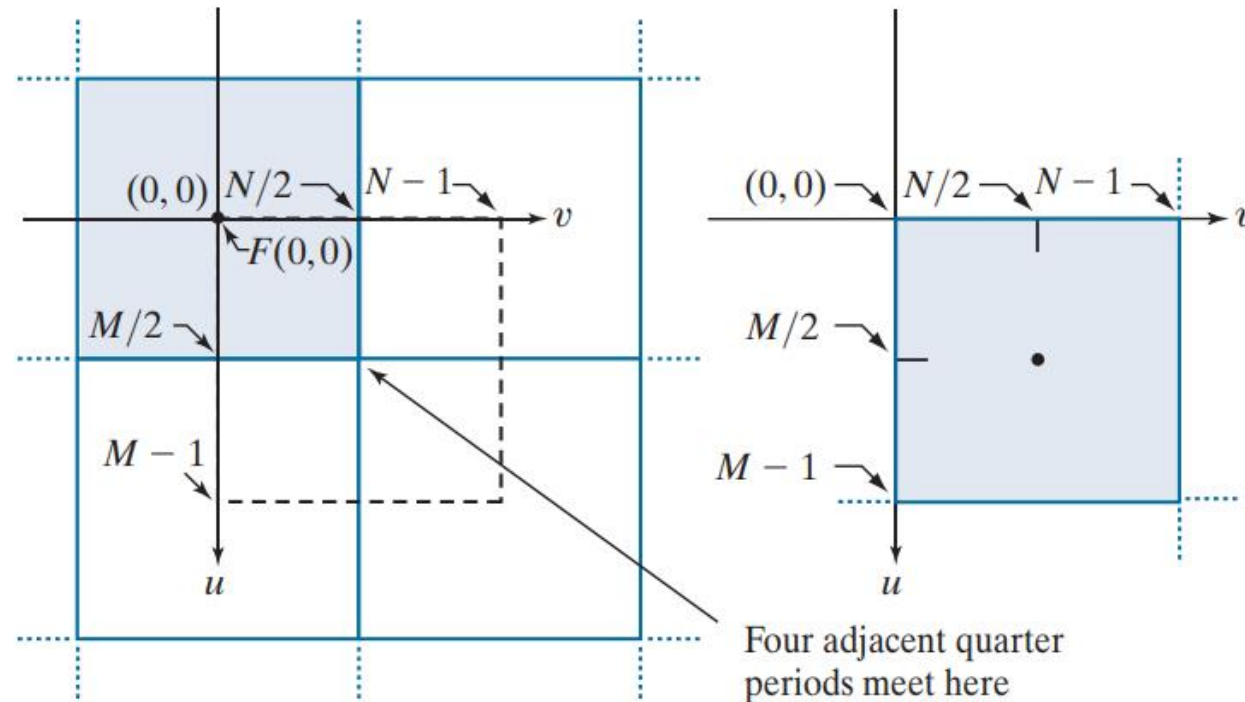
Let $u_0 = M/2$

$$f(x, y)(-1)^x \Leftrightarrow F(u - M/2)$$



Properties of the 2-D DFT

- Shifting



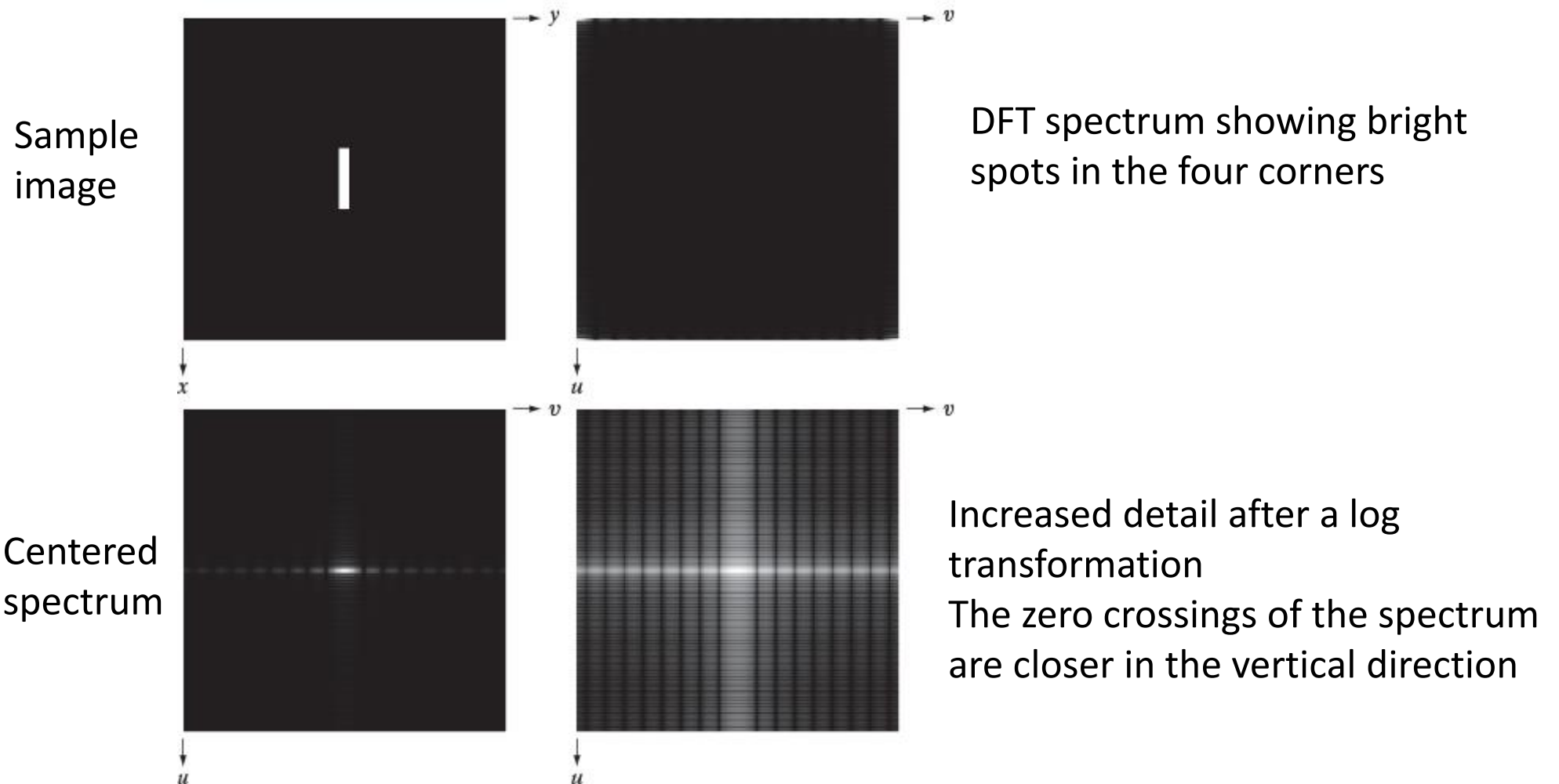
 = $M \times N$ data array computed by the DFT with $f(x, y)$ as input

 = $M \times N$ data array computed by the DFT with $f(x, y)(-1)^{x+y}$ as input

..... = Periods of the DFT

Properties of the 2-D DFT

Shift of the Origin Coordinates of the DFT



Properties of the 2-D DFT

- **Rotation**

By introducing the polar coordinates

$$x = r \cos \theta \quad y = r \sin \theta$$

$$u = \omega \cos \phi \quad v = \omega \sin \phi$$

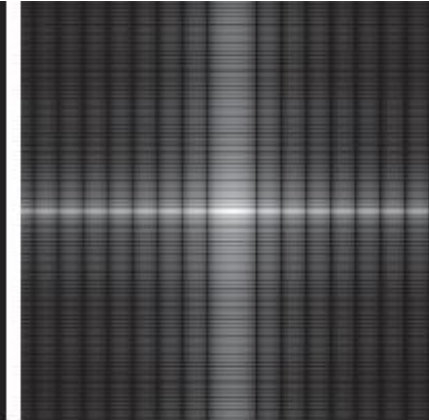
we have

$$f(r, \theta + \theta_0) \Leftrightarrow F(\omega, \phi + \theta_0)$$

sample
image



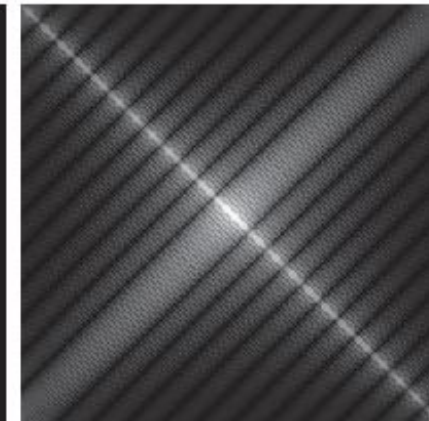
Corresponding
spectrum



rotated



Identical
rectangle



Properties of the 2-D DFT

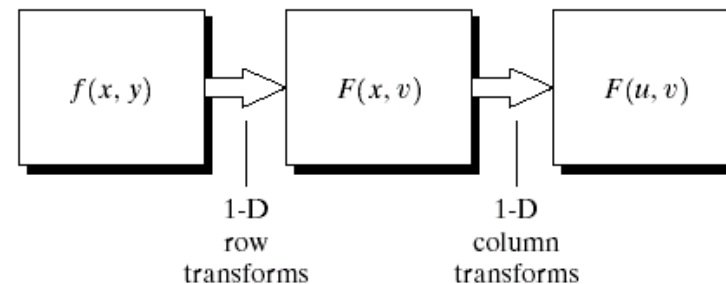
- Separability**

The DFT in can be expressed in the separable form

$$\begin{aligned} F(u, v) &= \frac{1}{M} \sum_{x=0}^{M-1} e^{-j2\pi ux/M} \frac{1}{N} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi vy/N} \\ &= \frac{1}{M} \sum_{x=0}^{M-1} F(x, v) e^{-j2\pi ux/M} \end{aligned}$$

where,

$$F(x, v) = \frac{1}{N} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi vy/N}$$



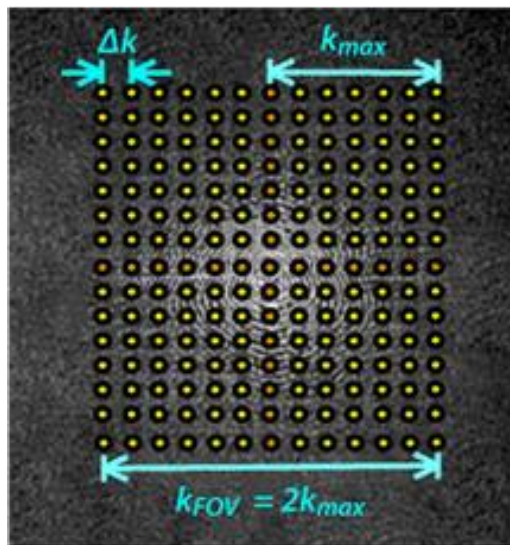
The same comments hold if we reverse the order of computation: columns first, followed by rows.

Properties of the 2-D DFT

- Relationships Between Spatial and Frequency Intervals**

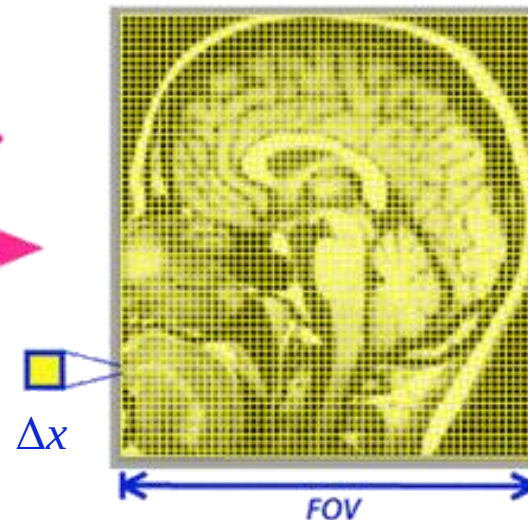
As in the 1-D case, there are the following relationships between samples in the spatial and frequency domains:

$$\Delta u = \frac{1}{M\Delta x} \quad \text{and} \quad \Delta v = \frac{1}{N\Delta y}$$



$$\Delta k = 1/\text{FOV}$$

FT



$$\Delta x = 1/k_{\text{FOV}}$$

The defined field-of view (FOV) and pixel width (Δx) determine the number of digitized samples in k -space

Properties of the 2-D DFT

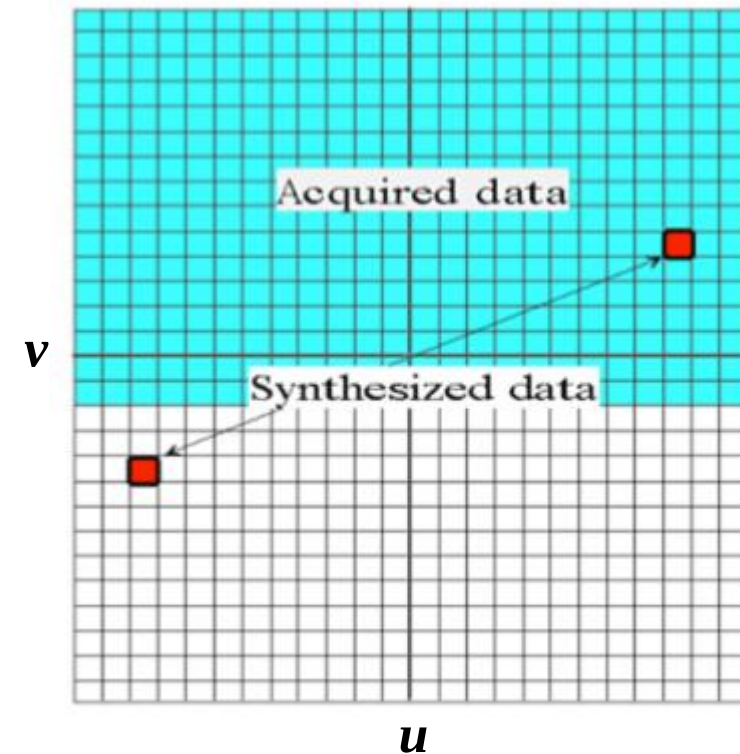
- **Conjugate symmetry**

If $f(x, y)$ is real (an image is naturally is), its FT is conjugate symmetric:

$$F(u, v) = F^*(-u, -v)$$

and its spectrum is symmetric:

$$|F(u, v)| = |F(-u, -v)|$$



Properties of the 2-D DFT

- **Convolution theorem**

The spatial filters and their frequency domain counterparts are linked by the *convolution theorem*.

$$f(x, y) * h(x, y) \Leftrightarrow F(u, v)H(u, v)$$

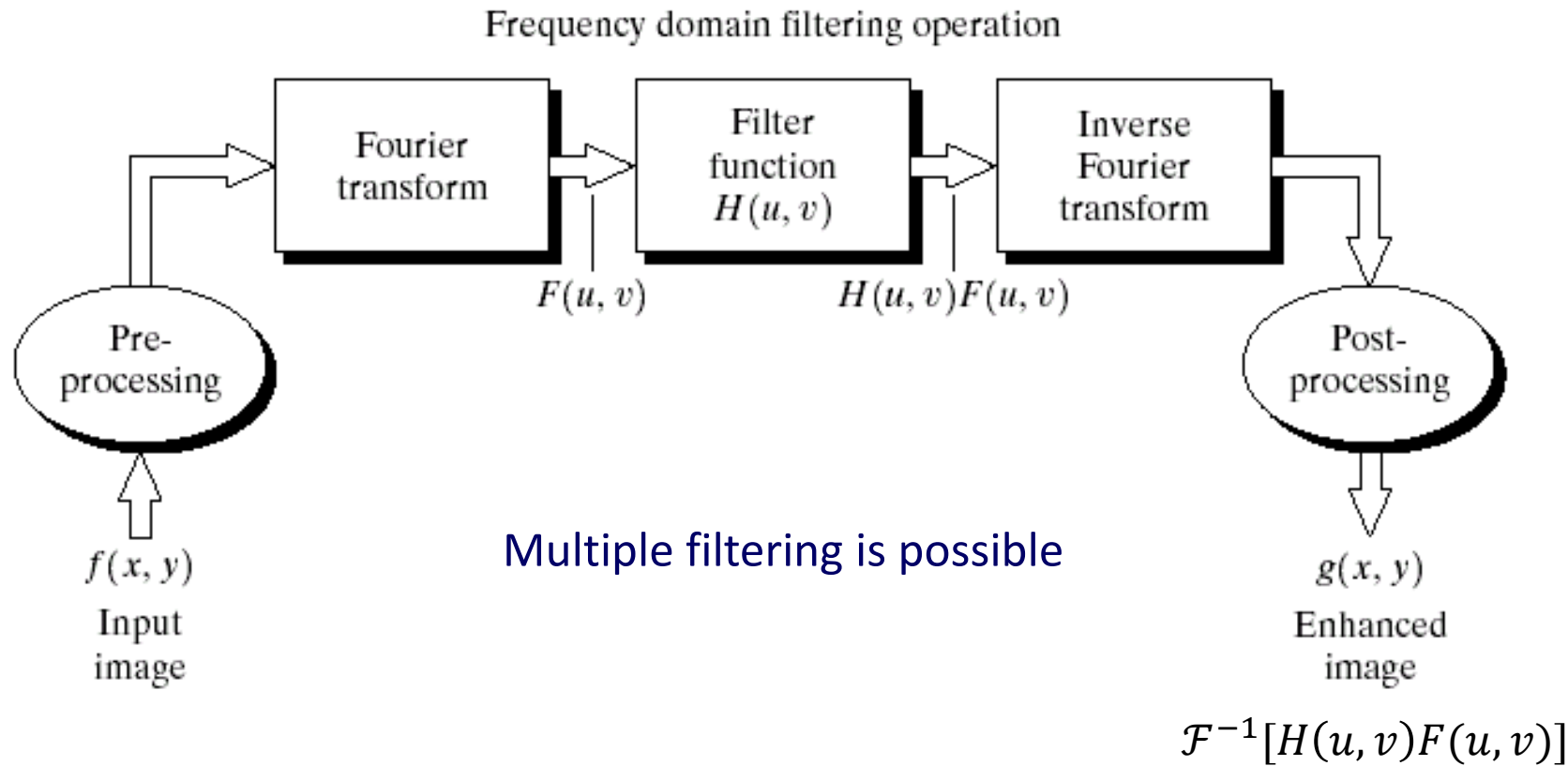
$$f(x, y)h(x, y) \Leftrightarrow F(u, v) * H(u, v)$$

Thus, the filters in the spatial and frequency domains constitute a FT pair.

In frequency domain, the time complexity of operate is lower.

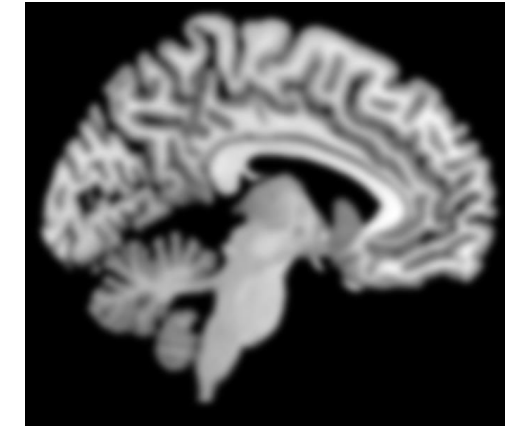
- Number of multiply in spatial: $T \sim O(M^2 K^2)$, M is the figure size and K is the kernel size.
- Number of multiply in frequency: $T \sim O(M^2)$, M is the figure size .

2D-DFT convolution theorem

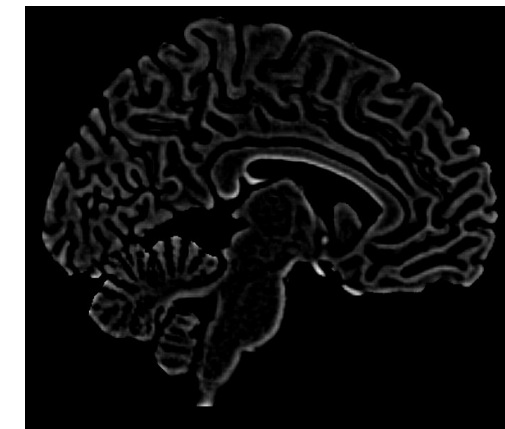


2D-DFT convolution example

blurred

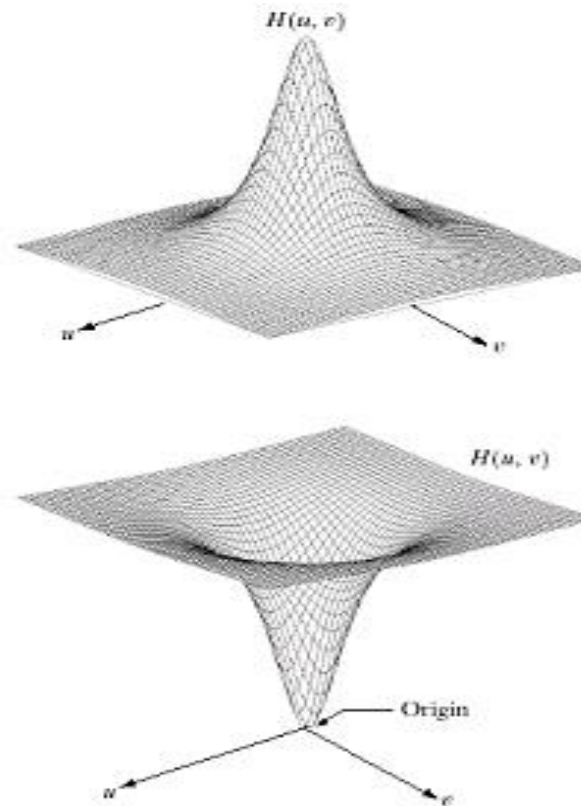


sharpened



Lowpass
filter

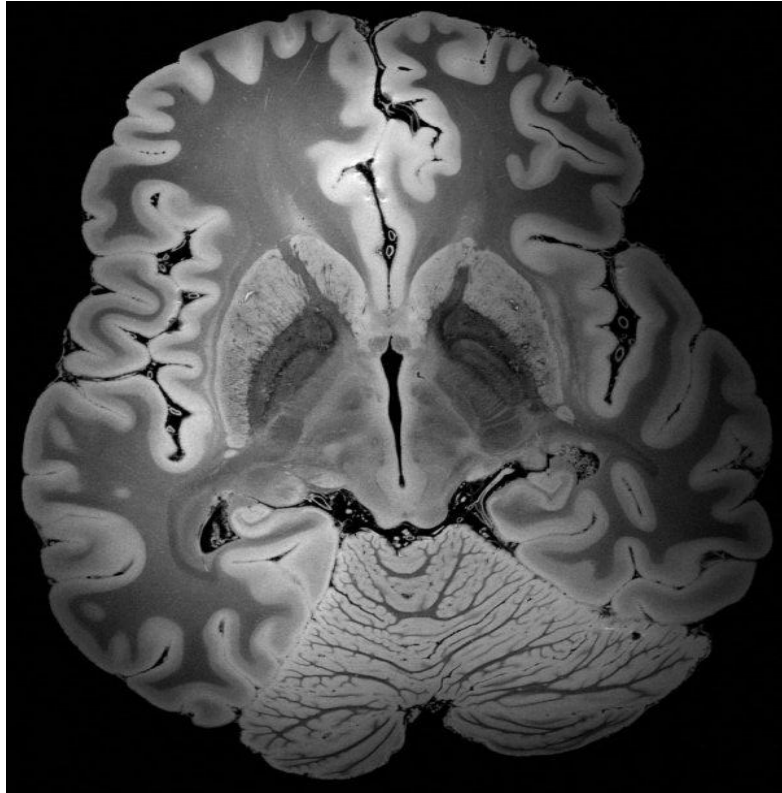
Highpass
filter



2-D DFT Summary

- 2D Fourier Transform
 - *Spatial domain to frequency domain*
- 2D Sampling and Aliasing
 - *2-D Nyquist criterion*
- 2-D DFT pairs between signal $f(x, y)$ and $F(u, v)$
 - The image is discrete 2-D function
- 2-D DFT properties
 - relationships between spatial and frequency intervals
 - conjugate symmetry
 - convolution theorem

The Fast Fourier Transform



2048 x 2048

- Directly perform DFT or IDFT
17 trillion computations



- FFT 92 million computations

Simplified

Speed up!

The Fast Fourier Transform

- In derivations of the FFT, we express

$$F(u) = \sum_{x=0}^{M-1} f(x)e^{-j2\pi ux/M} \quad \text{as} \quad F(u) = \sum_{x=0}^{M-1} f(x)W_M^{ux} \quad \text{where, } W_M = e^{-j2\pi/M}$$

- Successive-doubling method:** assume the number of samples is an integer power of 2:

$$M = 2^n \Rightarrow M = 2K \Rightarrow F(u) = \underbrace{\sum_{x=0}^{K-1} f(2x)W_{2K}^{u(2x)}}_{F_{\text{even}}(u)} + \underbrace{\sum_{x=0}^{K-1} f(2x+1)W_{2K}^{u(2x+1)}}_{F_{\text{odd}}(u)}$$

$$\downarrow \quad W_{2K}^{2ux} = W_K^{ux}$$

$$F(u) = \sum_{x=0}^{K-1} f(2x)W_K^{ux} + \sum_{x=0}^{K-1} f(2x+1)W_K^{ux}W_{2K}^u$$

The Fast Fourier Transform

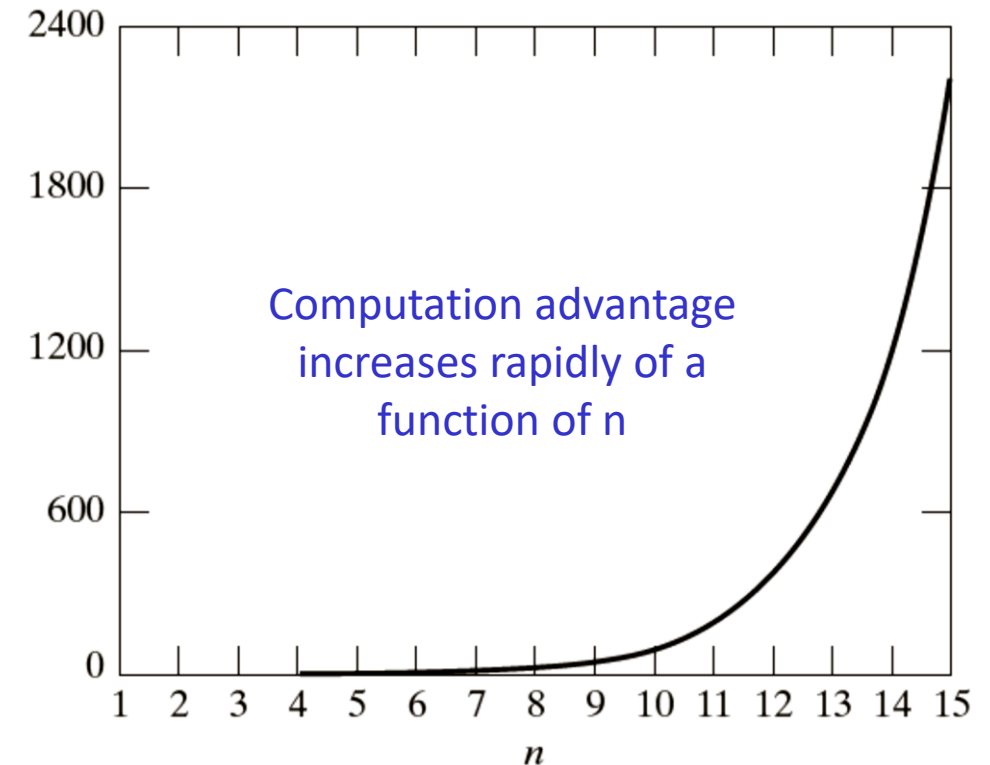
$$F(u) = \boxed{F_{\text{even}}(u)} + \boxed{F_{\text{odd}}(u)W_{2K}^u}$$

An M-point DFT can be computed by dividing the original expression into two parts.

The computational advantage of the FFT over a direct implementation of the 1-D DFT is defined as

$$C(M) = \frac{M^2}{M \log_2 M} = \frac{M}{\log_2 M}$$

$$\text{for } M = 2^n \quad C(n) = 2^n / n$$



Wavelet Transforms (Chapter 6)

- Introduction
- Wavelet Transforms in Multiresolution analysis
 - What is multiresolution analysis
 - Image pyramids
- Wavelet transforms
 - 1D Wavelet transform
 - 2D Wavelet transform

Wavelet Transforms - intro

- Transform = change of basis
 - Fourier Transform
 - Spherical Harmonic Transform
 - Wavelet Transform
 -
- See things differently

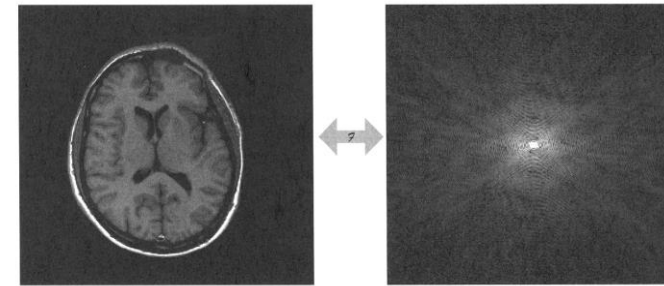
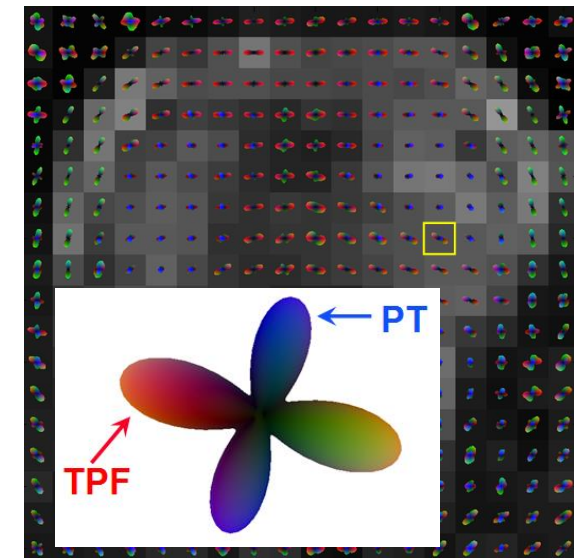
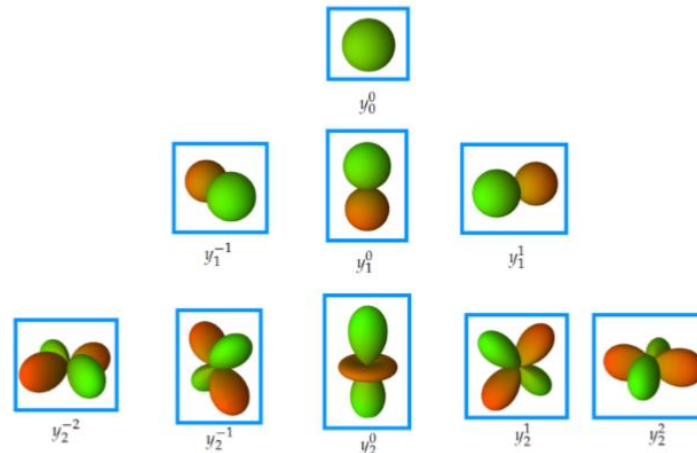


Image-space and k-space

Spherical Harmonics

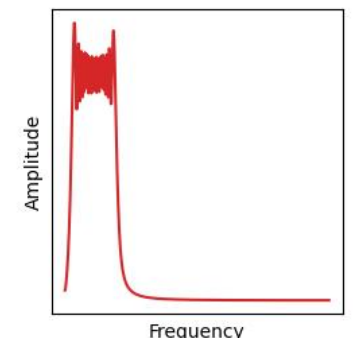
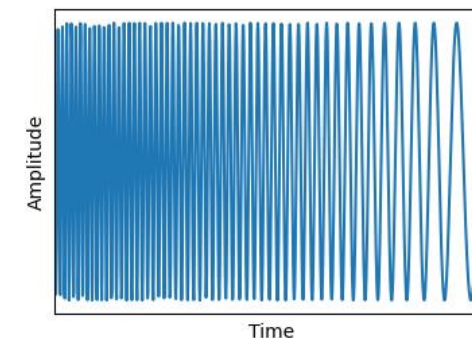
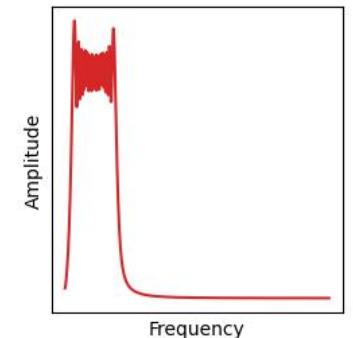
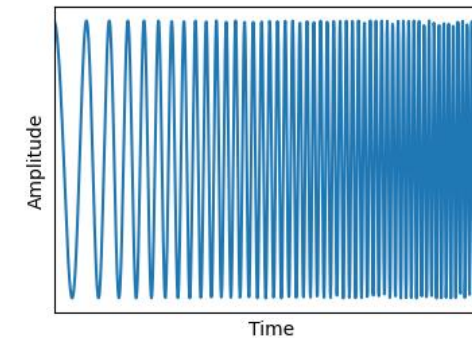
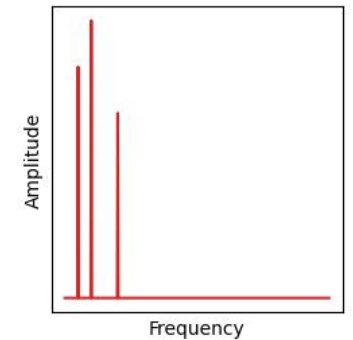
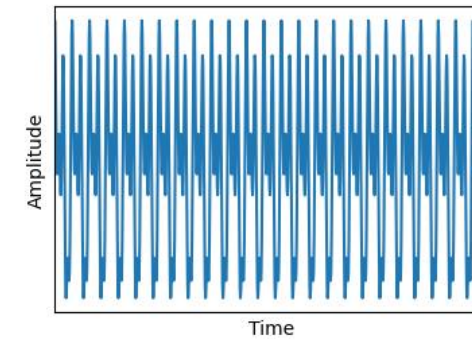


Wavelet Transforms - intro

- Motivation
 - Sometimes we care about **both frequency as well as time**

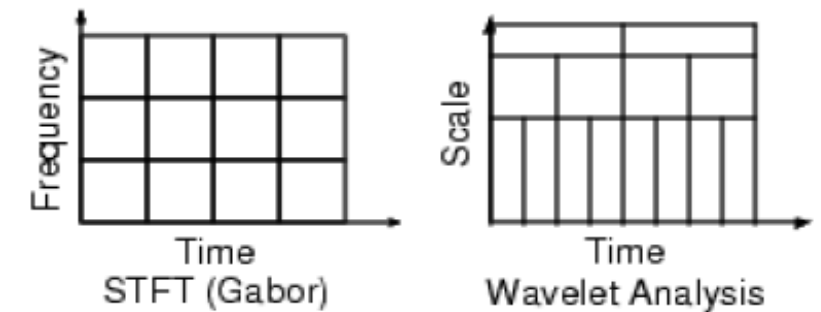
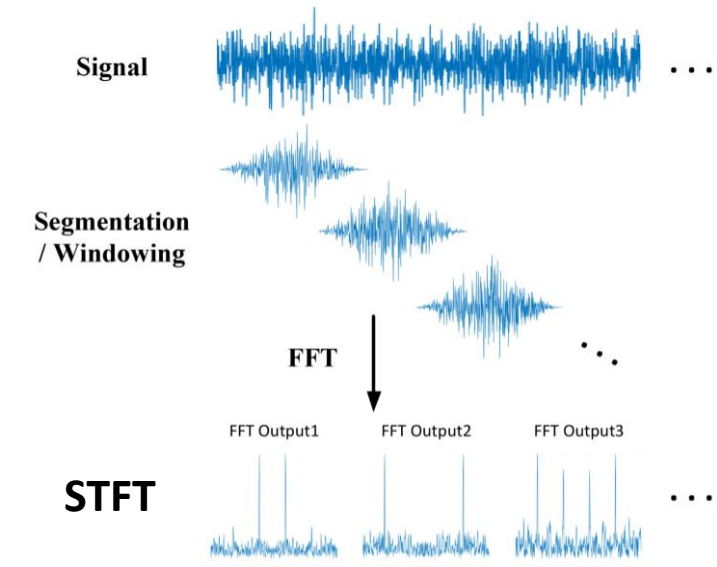


- Time domain operations – “when”
- Fourier domain operations – “how change (frequency)”
- Example
 - Fourier’s basis are sine/cosine functions
 - Fit **stationary signals** well
 - But Fourier’s basis can’t fit **time-varying frequency signals** well
 - Most real signals are non-stationary



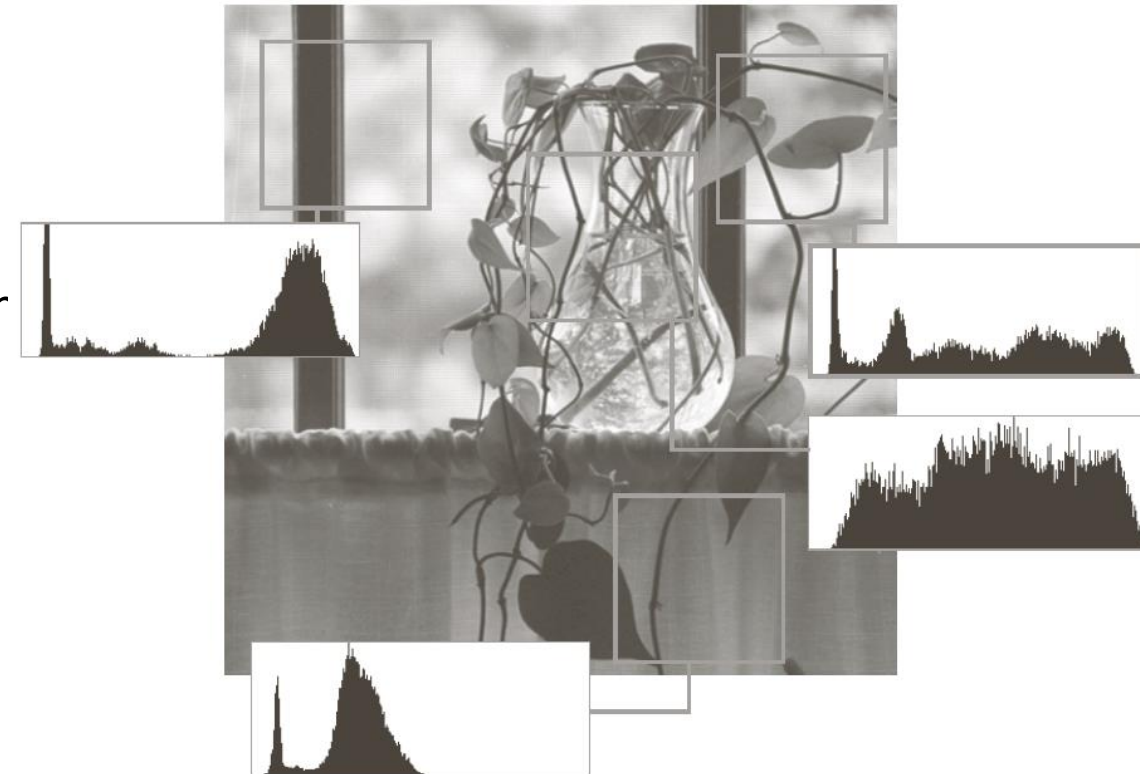
Wavelet Transforms - intro

- How to represent both time and frequency
 - **Based on Fourier:** short time Fourier Transform (**STFT**)
 - STFT: Fourier transform with **time windows** segment
 - Based on the other basis: **Wavelet Transforms (WT)**
- Differences between FT and WT
 - **Basis functions:** FT are sinusoids, WT are based on small waves, called wavelets
 - “Small” wave: **not stationary** to time
 - FT - only frequency information; WT – both frequency and temporal information



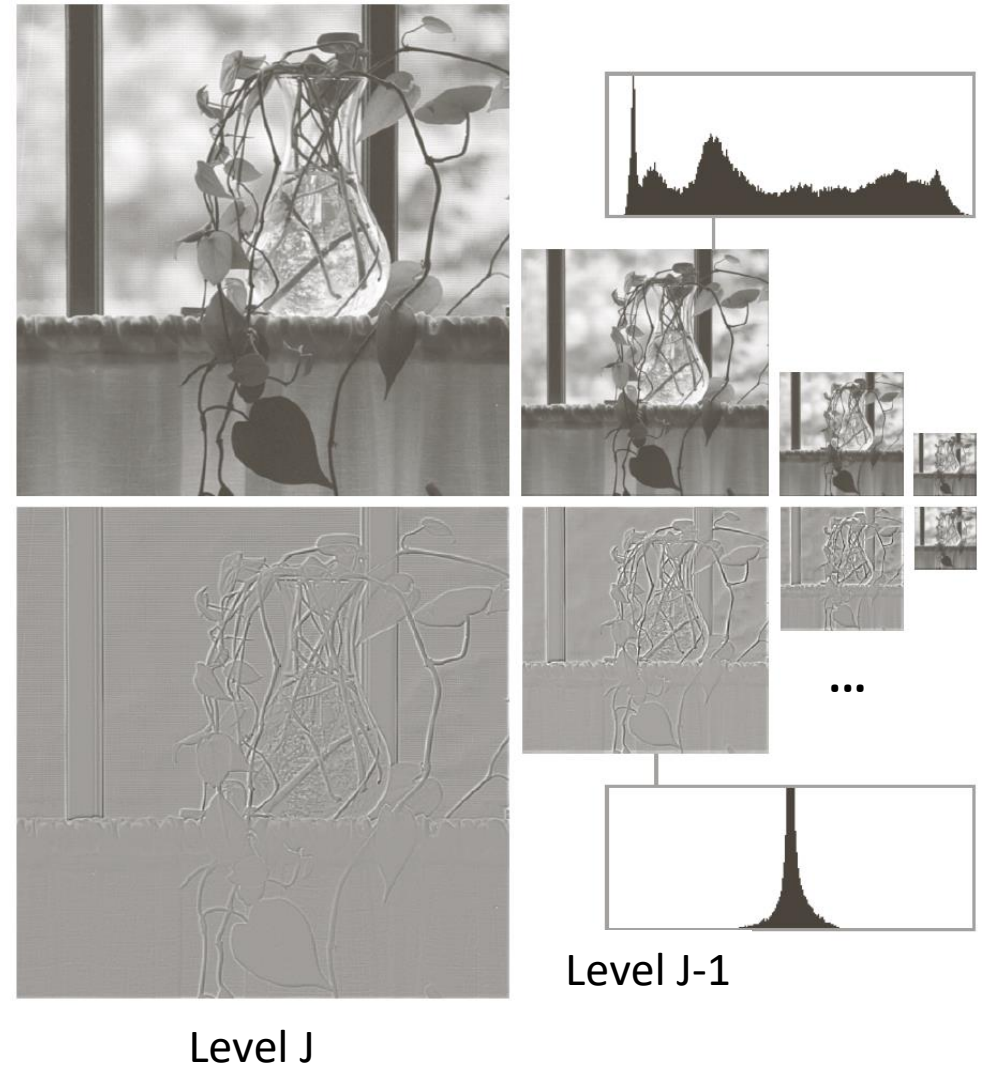
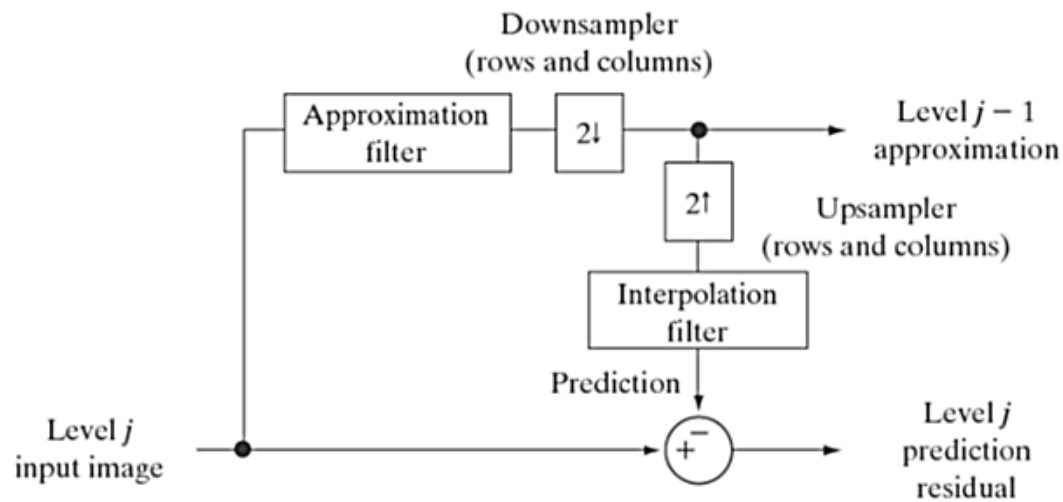
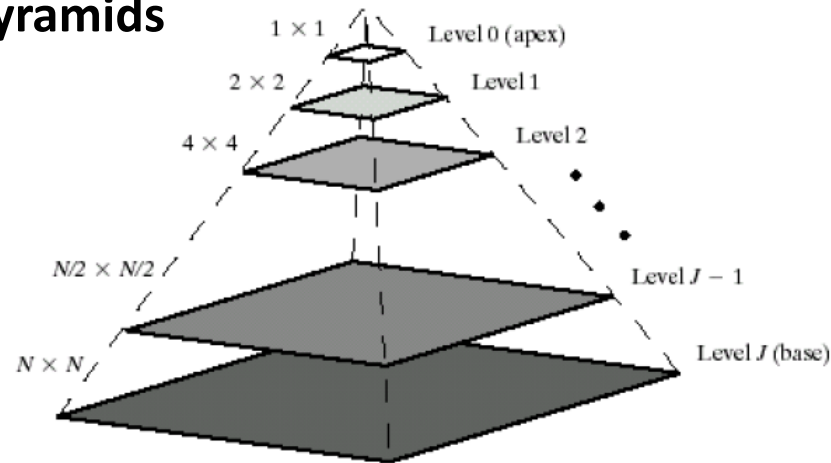
Wavelet Transforms in multiresolution analysis

- **What is multiresolution analysis**
 - If both small and large objects, or low and high contrast objects are present → need multiresolution
 - Local histogram variations – difficult to build a unified statistical model for the whole image



Wavelet Transforms in multiresolution analysis

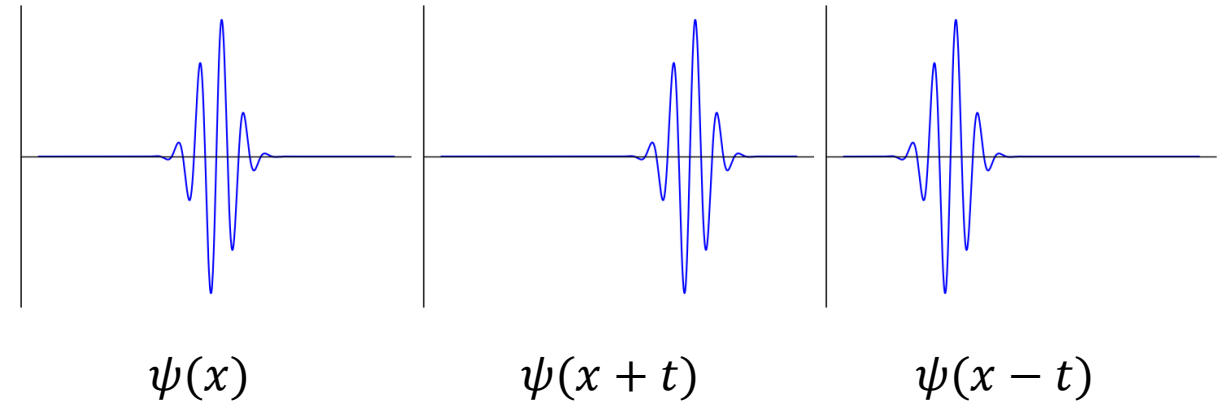
- Image pyramids



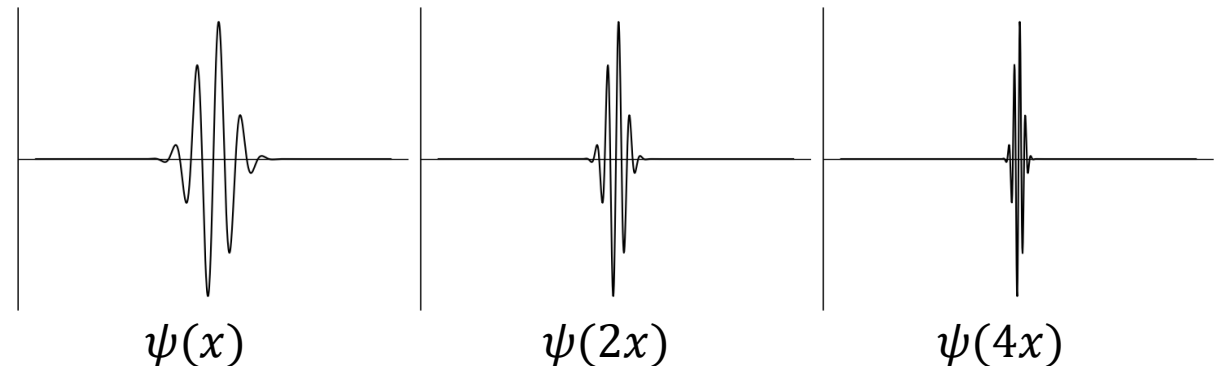
Wavelet Transforms (1D)

- **Wavelet transform in one dimension**
 - Continuous **W**avelet **T**ransform
 - Define a function $\psi(x)$ (Mother wavelet)
 - assume $\psi(x)$ band-limited and its DC component = 0 ($\int \psi(x)dx = 0$)
 - Create **scaled** and **shifted** versions of $\psi(x)$
 - **Scaling**: Change function's frequency
 - **Shifting**: Translation function in time
 - $\psi_{s,\tau}(x) = \frac{1}{\sqrt{s}} \psi\left(\frac{x-\tau}{s}\right)$

Shifting

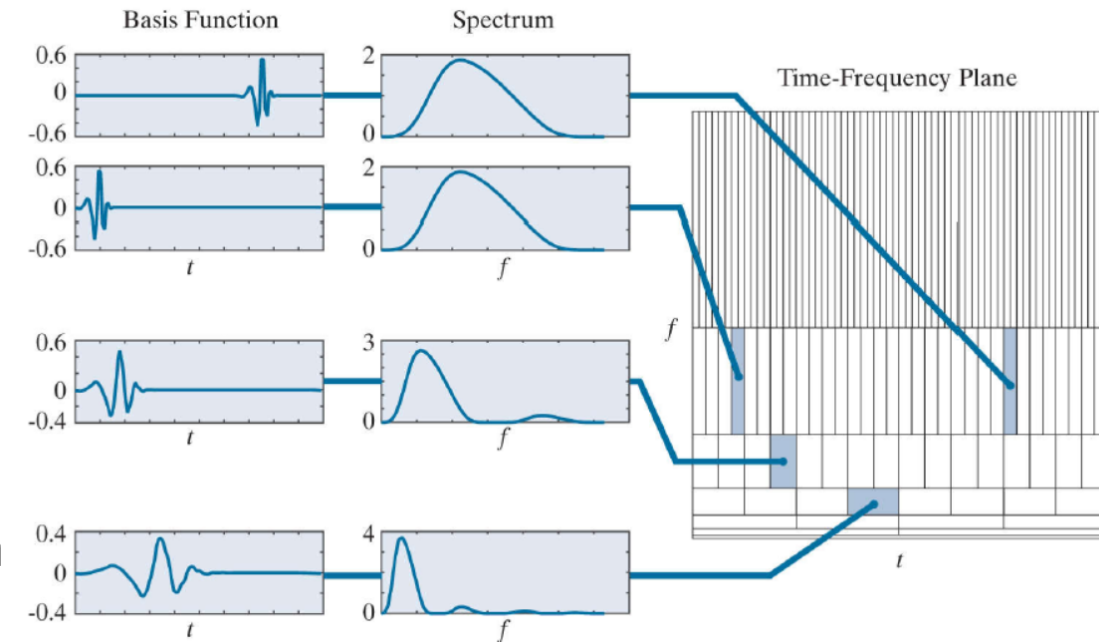


Scaling



Wavelet Transforms (1D)

- Wavelet transform in one dimension
 - Continuous **W**avelet **T**ransform
 - Define CWT of $f(x)$:
 - $W_{\psi}(s, \tau) = \int_{-\infty}^{\infty} f(x) \psi_{s, \tau}(x) dx$
 - The wavelet coefficients measure how closely correlated the wavelet is with each section (basis) of the signal



Wavelet Transforms (1D)

- Wavelet transform in one dimension

- Continuous Wavelet Transform

- $$\psi_{s,\tau}(x) = \frac{1}{\sqrt{s}} \psi\left(\frac{x-\tau}{s}\right)$$
- $$W_\psi(s, \tau) = \int_{-\infty}^{\infty} f(x) \psi_{s,\tau}(x) dx$$

- Inverse Transform

- $$f(x) = \frac{1}{C_\psi} \int_0^\infty \int_{-\infty}^{\infty} \frac{W_\psi(s, \tau) \psi_{s,\tau}(x)}{s^2} d\tau ds$$
- $$C_\psi = \int_{-\infty}^{\infty} \frac{|\Psi(\mu)|}{|\mu|} d\mu$$
- where $\Psi(\mu)$ is the Fourier transform of $\psi(x)$

- Discrete Wavelet Transform

- $$\psi_{j,k}(x) = 2^{\frac{j}{2}} \psi(2^j x - k)$$

- $$W_\varphi(j_0, k) = \frac{1}{\sqrt{M}} \sum_n f(n) \varphi_{j_0,k}(n)$$

← “Approximation” coefficients by scaling function

- $$W_\psi(j, k) = \frac{1}{\sqrt{M}} \sum_n f(n) \psi_{j,k}(n)$$

← “Detail” coefficients by wavelet function

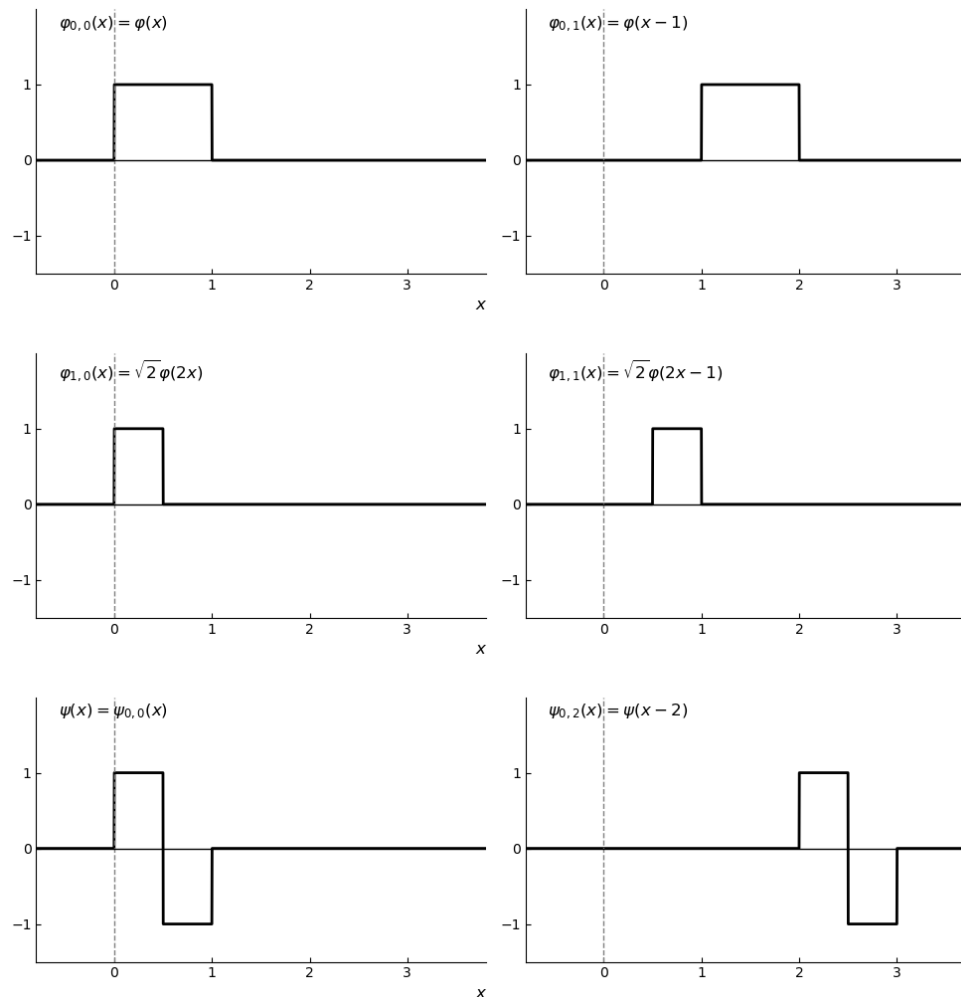
- Inverse Transform

- $$f(n) = \frac{1}{\sqrt{M}} \sum_k W_\varphi(j_0, k) \varphi_{j_0,k}(n) + \frac{1}{\sqrt{M}} \sum_j \sum_k W_\psi(j, k) \psi_{j,k}(n)$$
- where j_0 is an arbitrary starting scale, and $n = 0, 1, 2, \dots, M$

- For CWT, the $s \rightarrow \infty$ is the DC component
- For DWT, the DC component needs scaling function

Wavelet Transforms (1D)

- Example



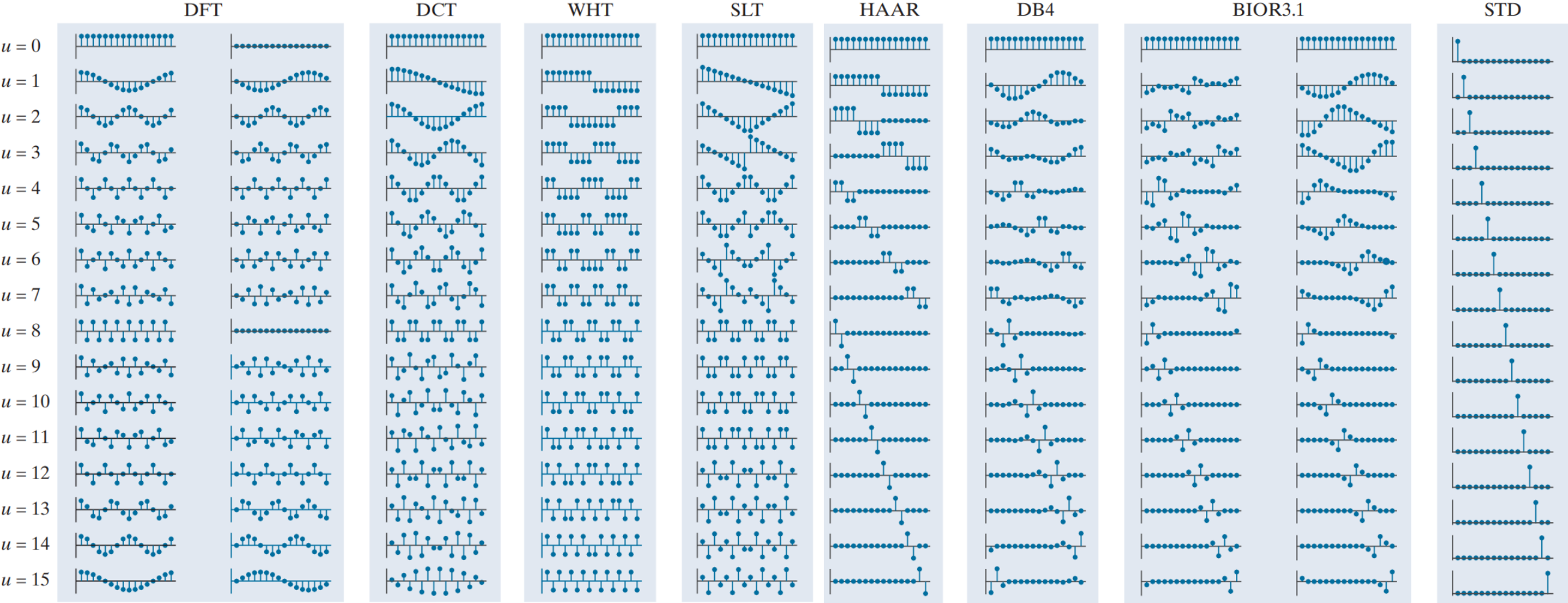
Haar scaling functions φ

- Father wavelet
- $DC \neq 0$
- For low frequency

Haar wavelet functions ψ

- Mother wavelet
- $DC = 0$
- For high frequency

Wavelet Transforms (1D)

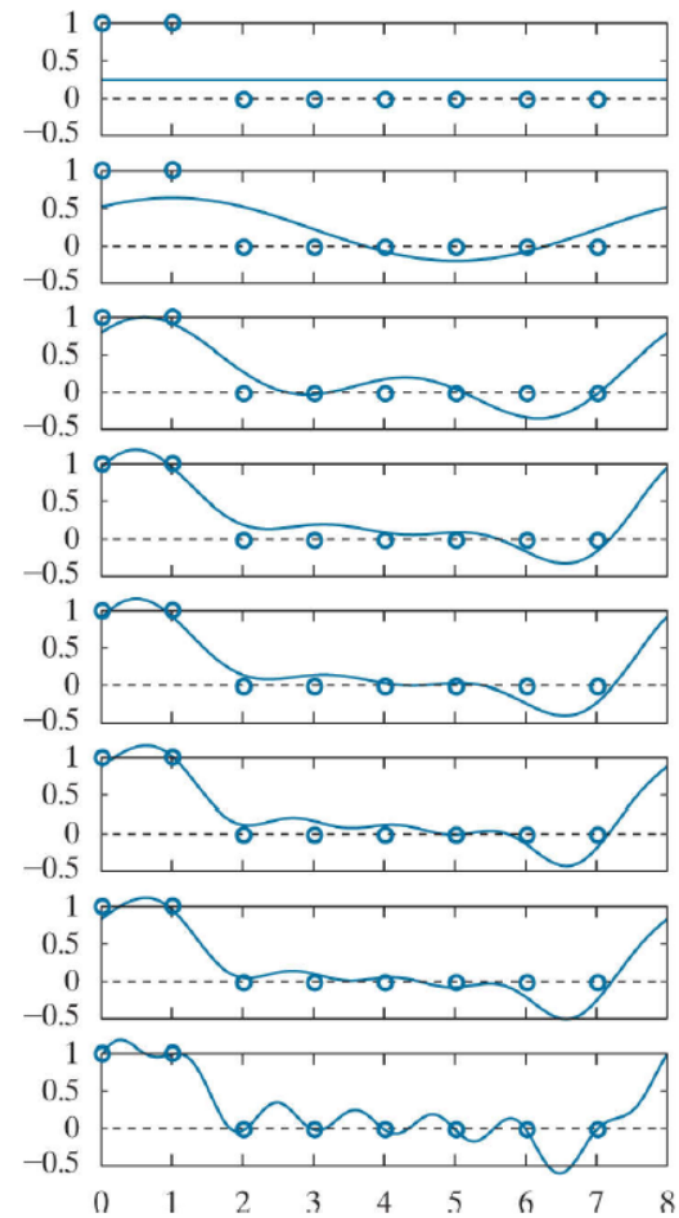


Stationary basis

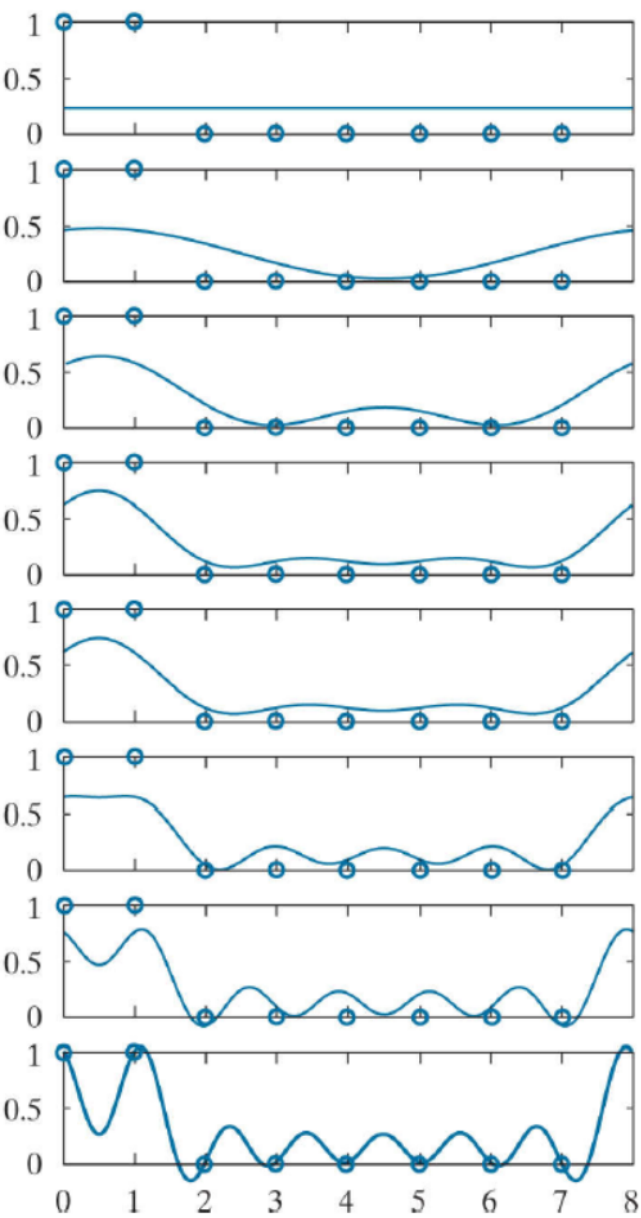
Not-stationary basis



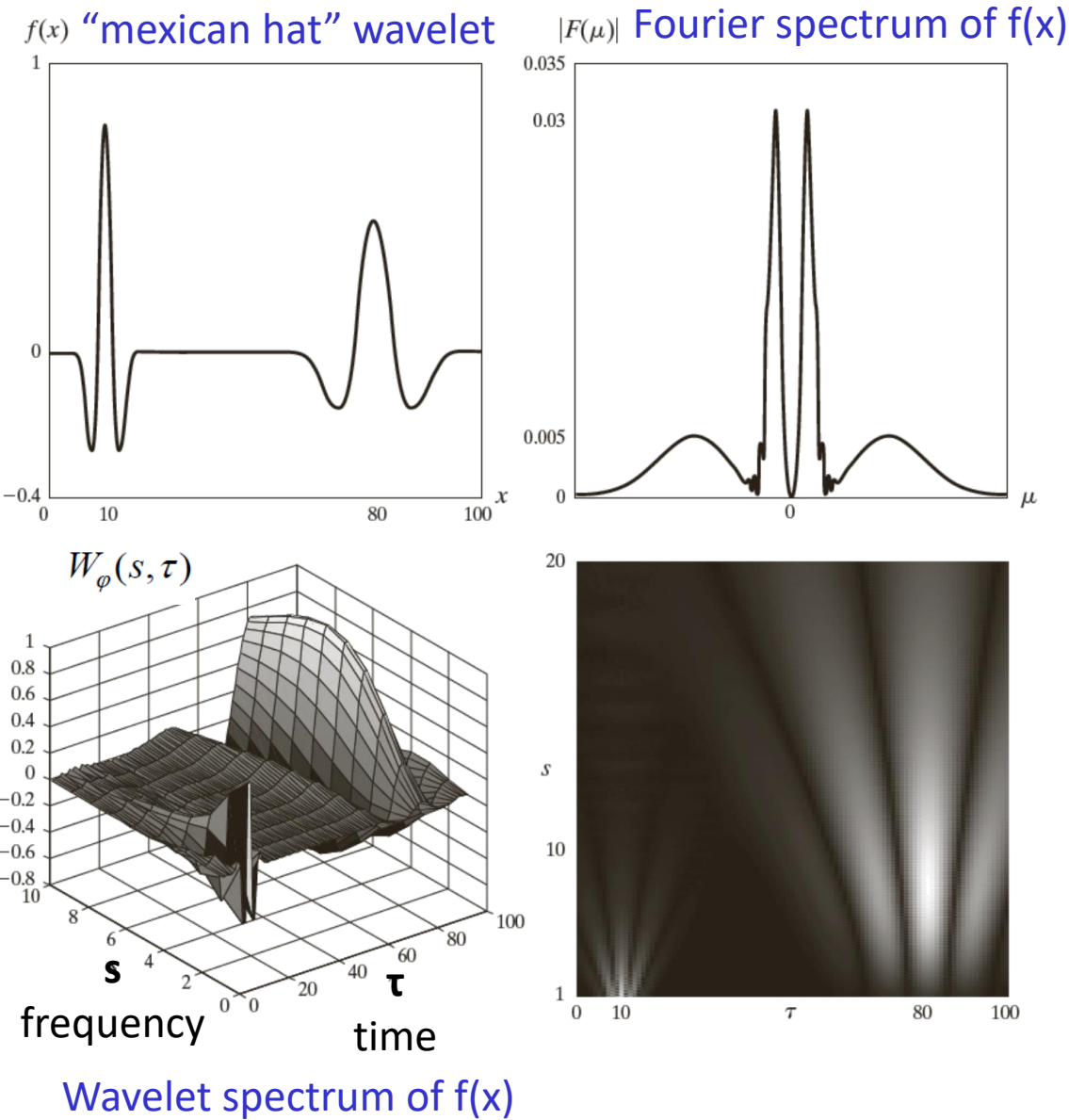
Discrete Hartley Transform (DHT)



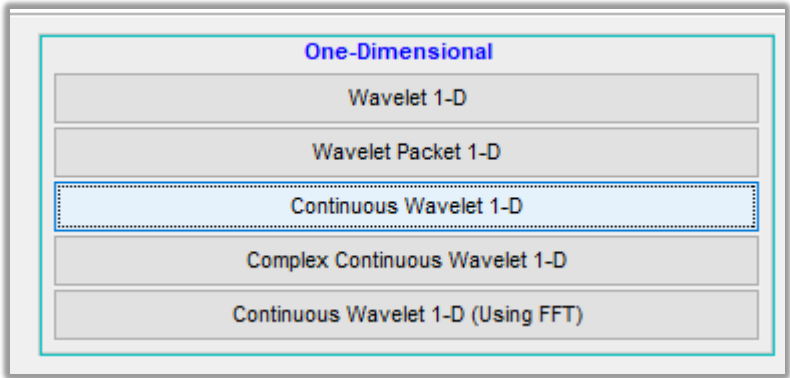
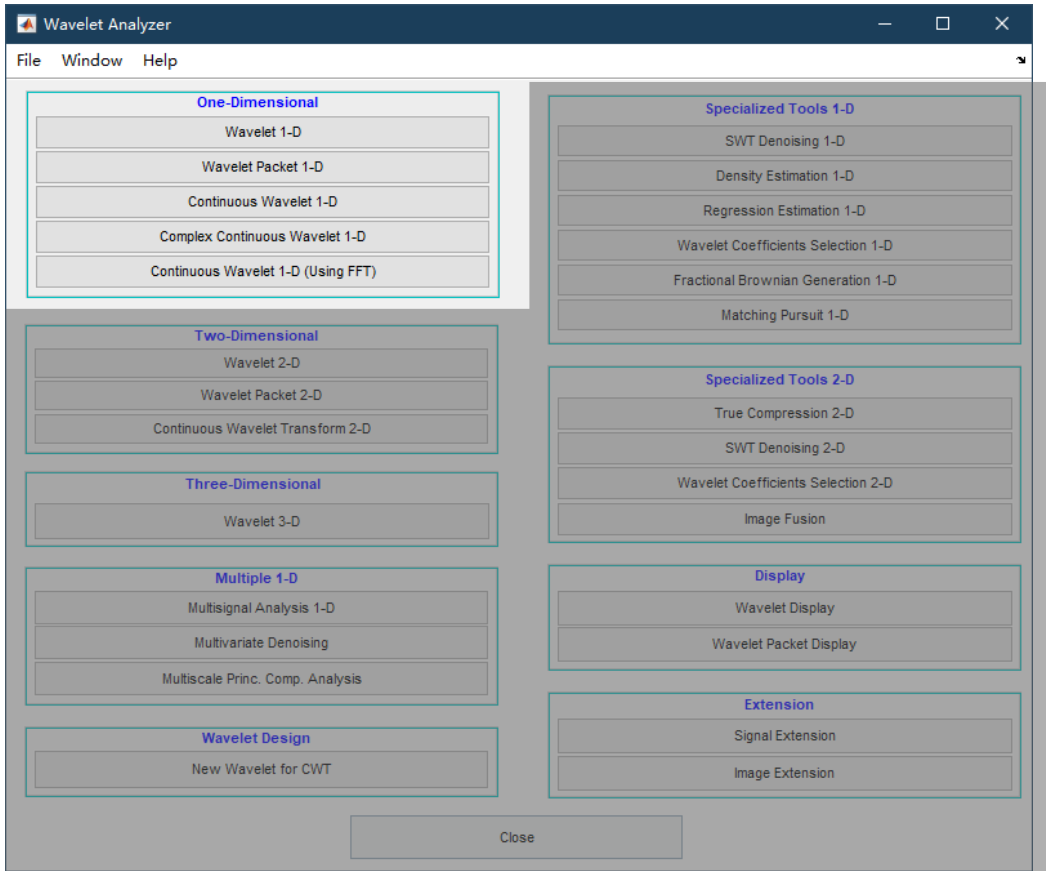
Discrete Fourier Transform (DFT)



- Example



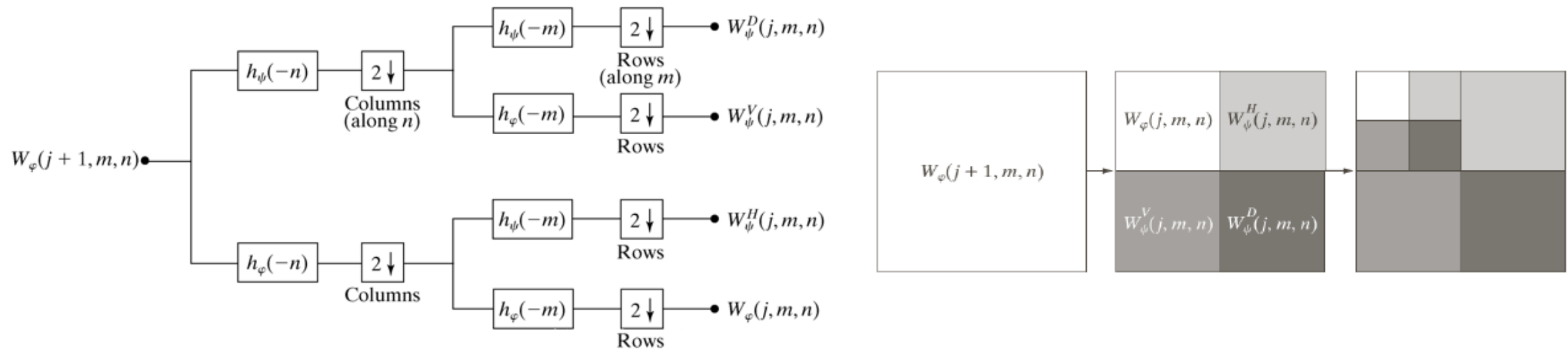
wavemenu in Matlab



2D Wavelet Transforms

- Wavelet transform in two dimension

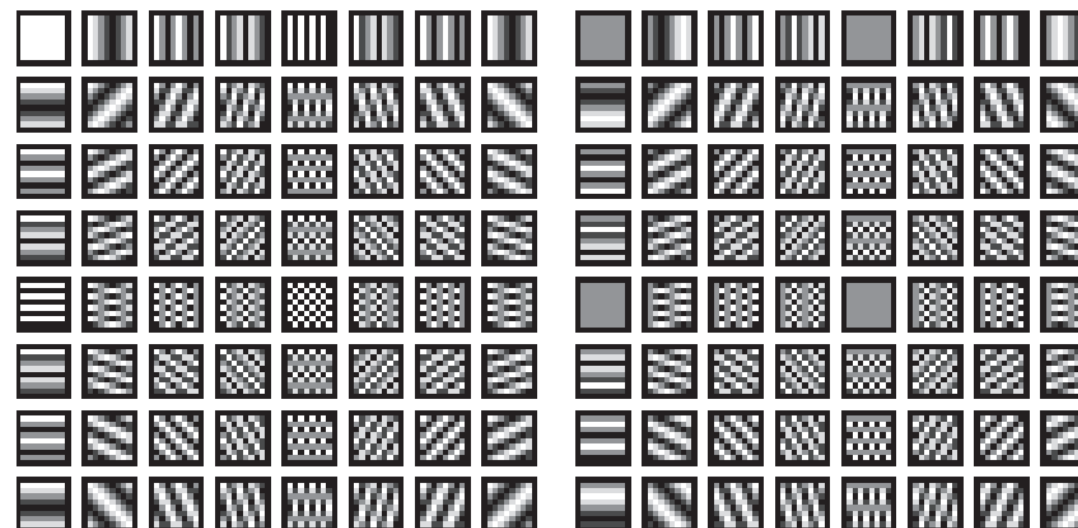
A two-dimensional **scaling function**, $\varphi(x, y)$, and three two-dimensional **wavelet functions**, $\psi^H(x, y)$, $\psi^V(x, y)$, $\psi^D(x, y)$, are required.



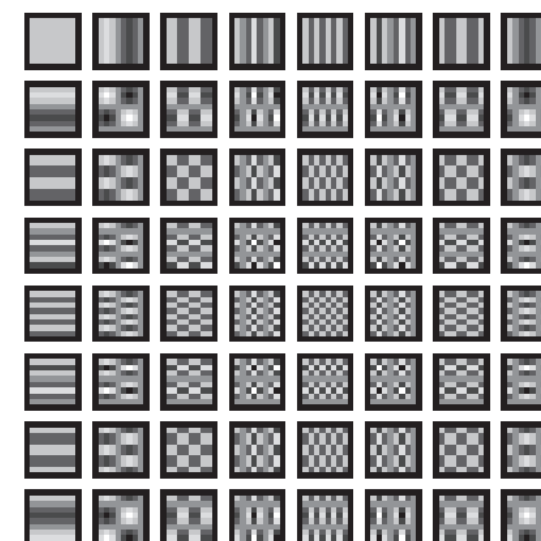
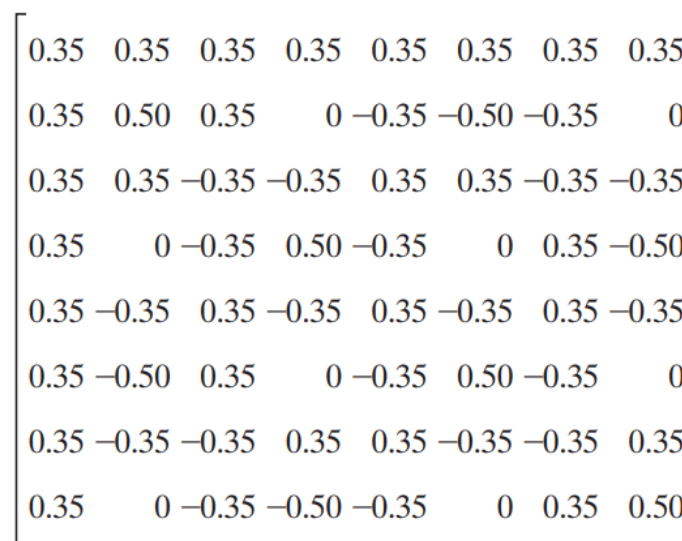
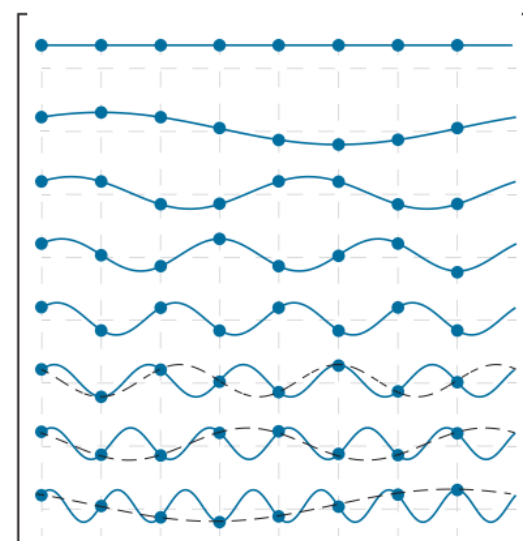
Wavelet Transforms

Discrete Fourier
transform

$$\frac{1}{\sqrt{8}} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & \omega & -j & -j\omega & -1 & -\omega & j & j\omega \\ 1 & -j & -1 & j & 1 & -j & -1 & j \\ 1 & -j\omega & j & \omega & -1 & j\omega & -j & -\omega \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -\omega & -j & j\omega & -1 & \omega & j & -j\omega \\ 1 & j & -1 & -j & 1 & j & -1 & -j \\ 1 & j\omega & j & -\omega & -1 & -j\omega & -j & \omega \end{bmatrix}$$

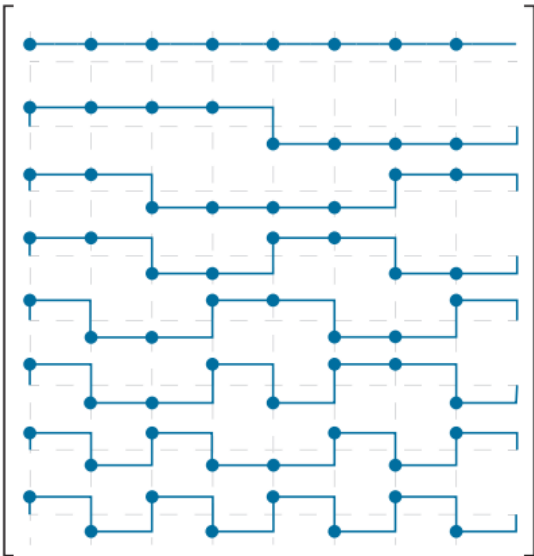


Discrete Hartley
transform

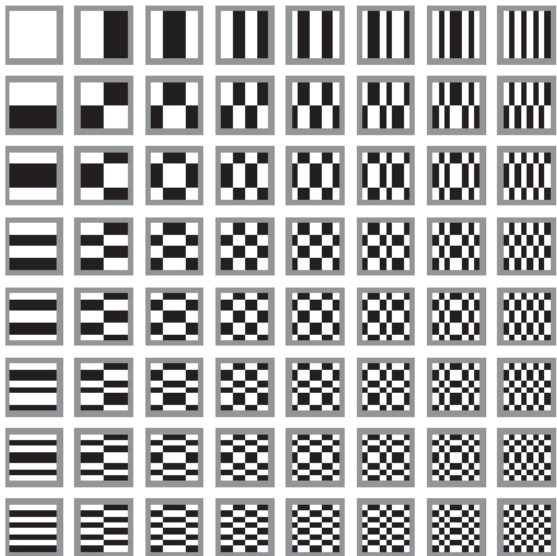


Wavelet Transforms

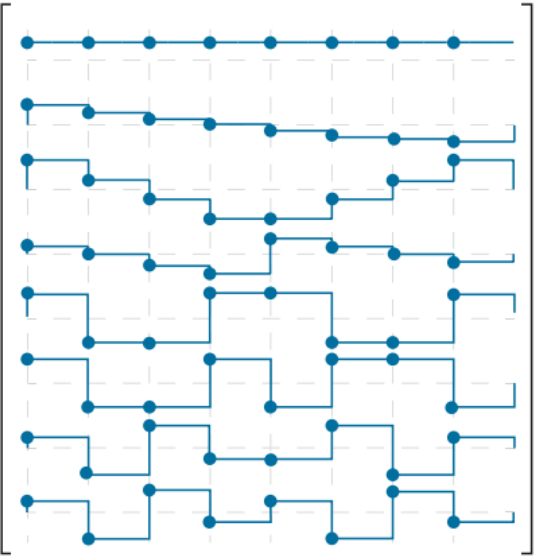
Hadamard transform



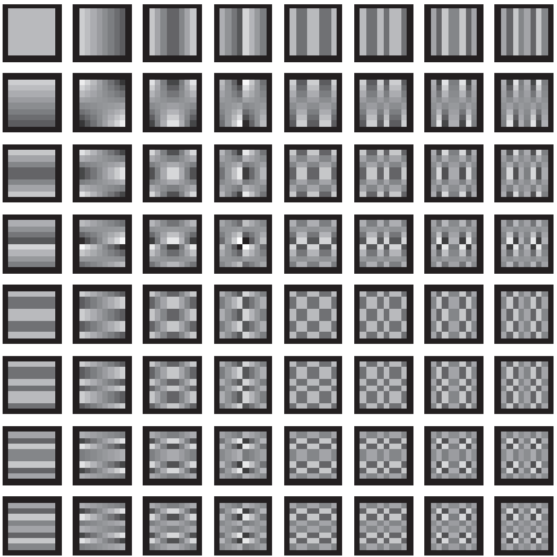
0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35
0.35	0.35	0.35	0.35	-0.35	-0.35	-0.35	-0.35
0.35	0.35	-0.35	-0.35	-0.35	-0.35	0.35	0.35
0.35	0.35	-0.35	-0.35	0.35	0.35	-0.35	-0.35
0.35	-0.35	-0.35	0.35	0.35	-0.35	-0.35	0.35
0.35	-0.35	-0.35	0.35	-0.35	0.35	0.35	-0.35
0.35	-0.35	0.35	-0.35	-0.35	0.35	-0.35	0.35
0.35	-0.35	0.35	-0.35	0.35	-0.35	0.35	-0.35



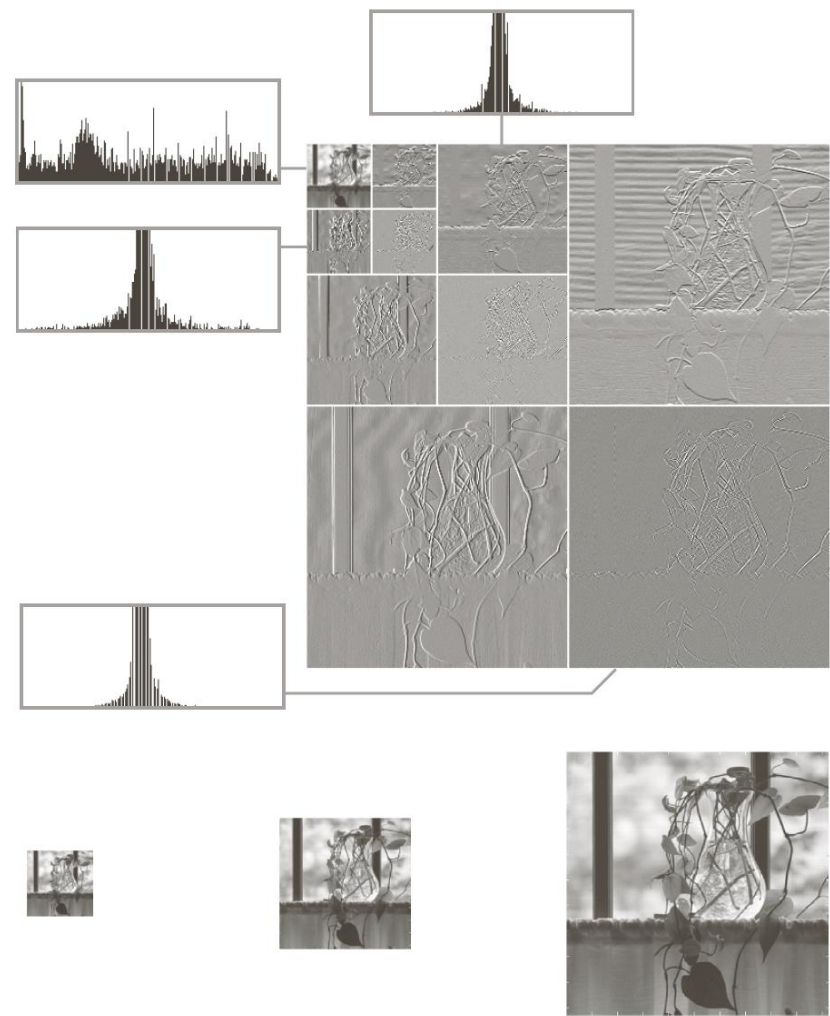
Slant transform



0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35
0.54	0.39	0.23	0.08	-0.08	-0.23	-0.39	-0.54
0.47	0.16	-0.16	-0.47	-0.47	-0.16	0.16	0.47
0.24	-0.04	-0.31	-0.59	0.59	0.31	0.04	-0.24
0.35	-0.35	-0.35	0.35	0.35	-0.35	-0.35	0.35
0.35	-0.35	-0.35	0.35	-0.35	0.35	0.35	-0.35
0.16	-0.47	0.47	-0.16	-0.16	0.47	-0.47	0.16
0.16	-0.47	0.47	-0.16	0.16	-0.47	0.47	-0.16

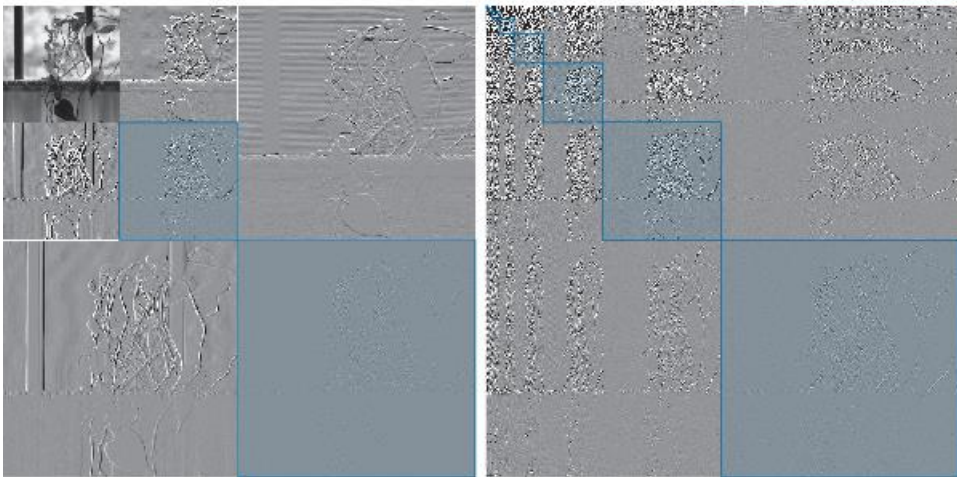
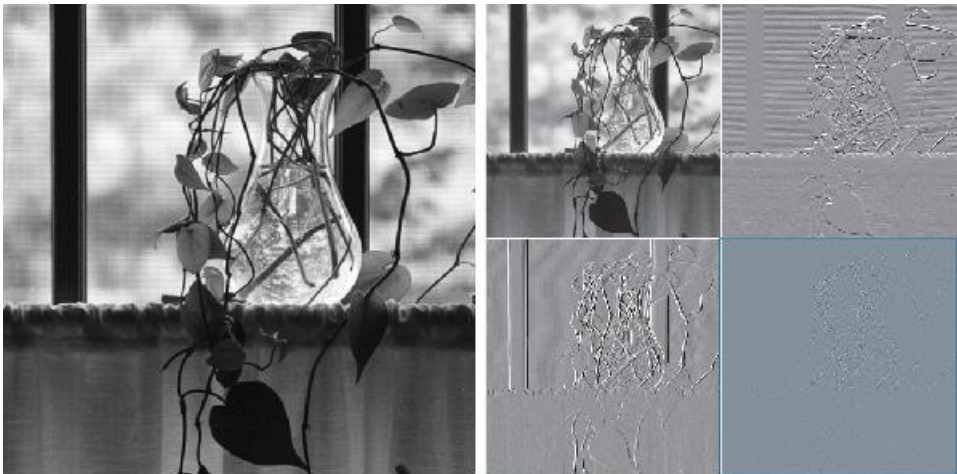


Wavelet Transforms



A DWT using Haar H_2 basis functions and its local variable histogram

One scale FWT

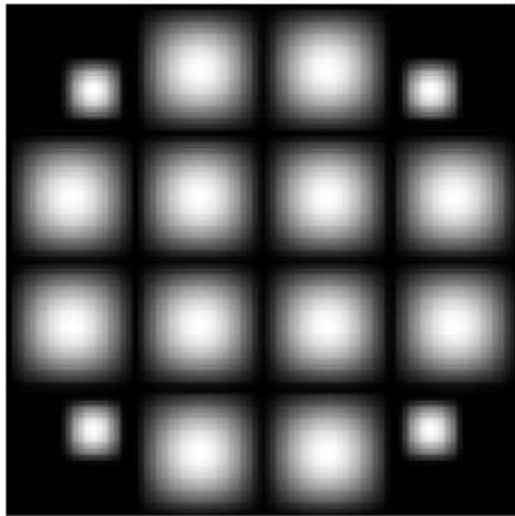


Two scale FWT

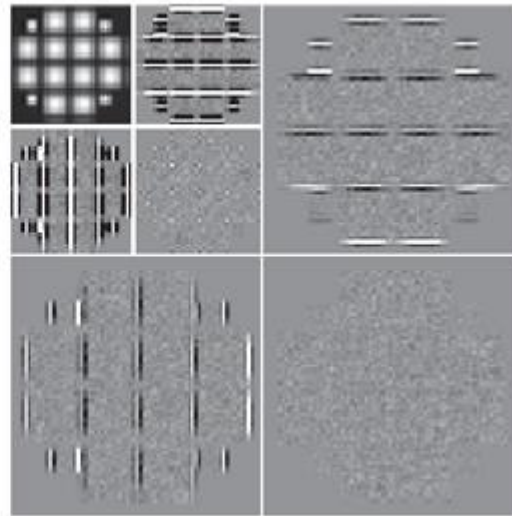
Haar transformation

Wavelet Transforms

- Application
 - Edge detection

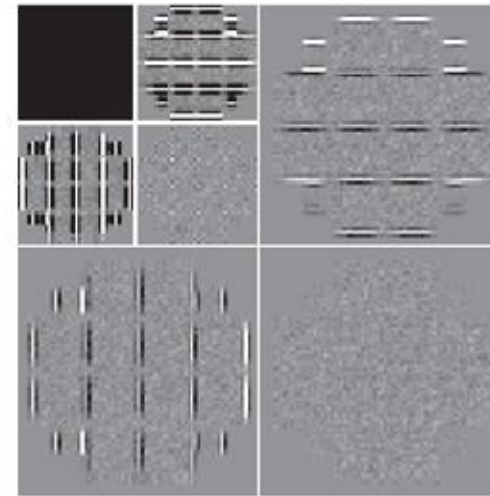


Original Image

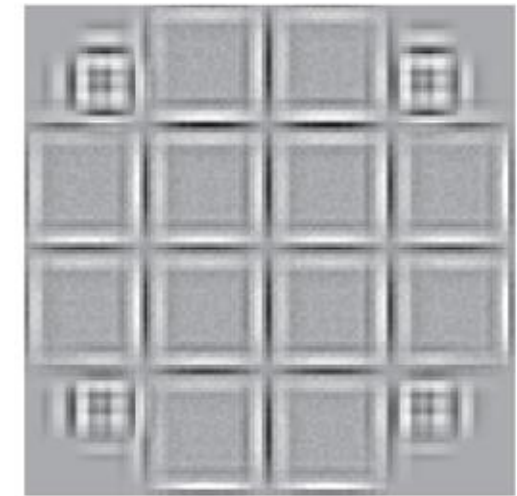


Two scale DWT with 4th order symlets

Set approximation to 0



Reconstructions
(Inverse WT)



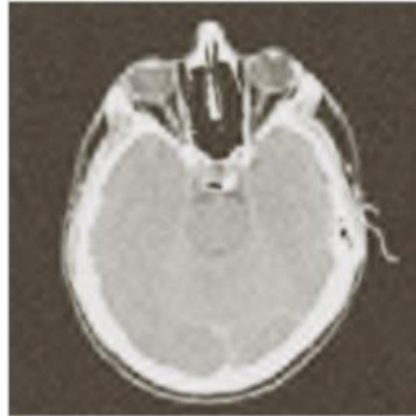
Wavelet Transforms

- **Application**
 - Denoise

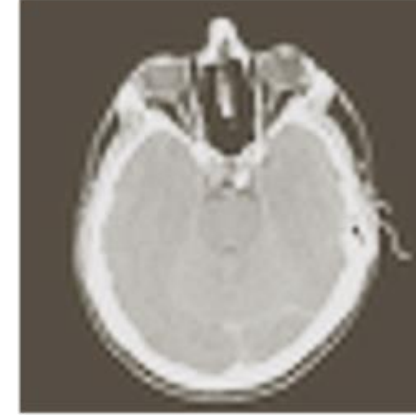
Original noisy image



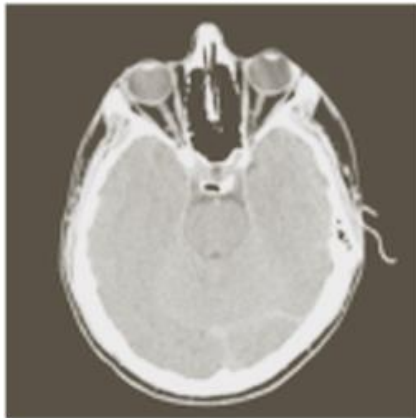
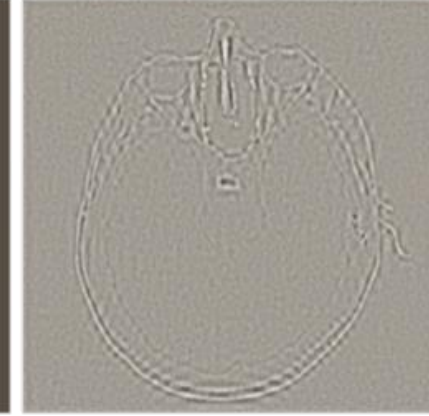
Reconstruction1



Reconstruction2



Original - R2



Reconstruction3



Original - R3

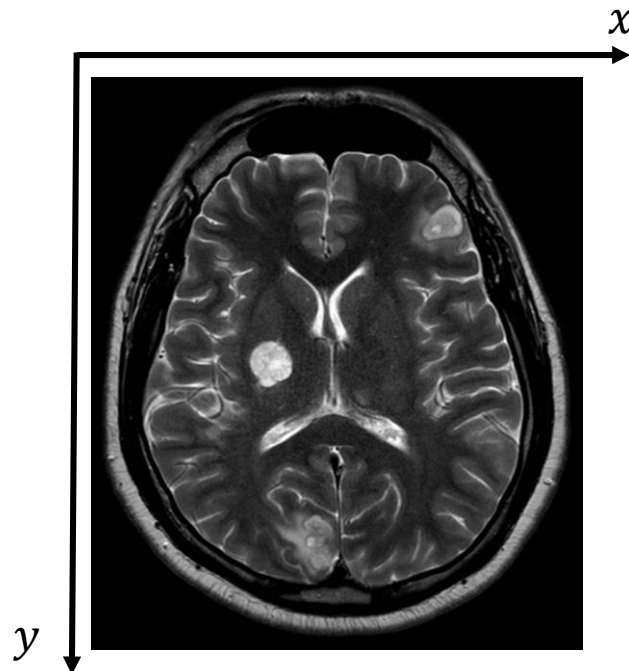
R1, R2, and R3 are various reconstructions after thresholding the detail coefficients

Fourier Transform in Digital Image

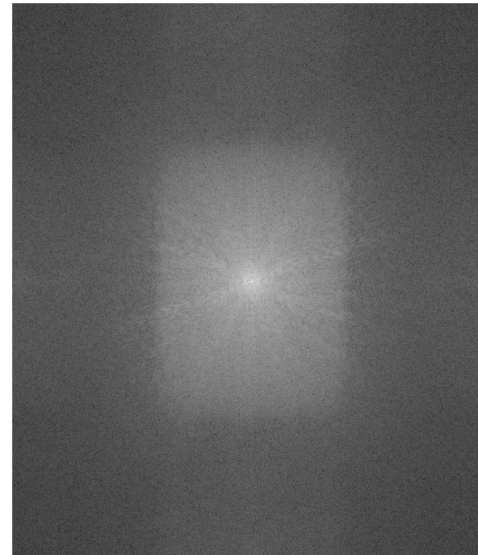
- Fourier Transformation
 - **2D Fourier Transform**
 - **2D Discrete Fourier Transform**
 - Sampling and Aliasing
 - Properties
 - Fast Fourier Transform
- Wavelet Transform

Homework 1

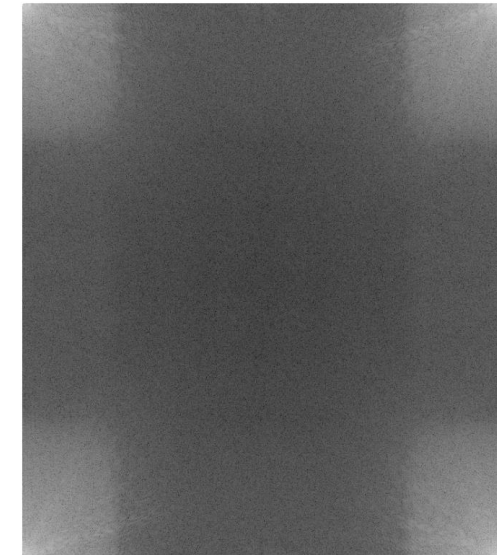
- For an image with size $M \times N$, prove that multiplying the image with $(-1)^{x+y}$ and applying Fourier transform, the DC component of the result frequency spectrum will in the center $[\frac{M}{2}, \frac{N}{2}]$.
- What will happen to the spectrum if we replace the $(-1)^{x+y}$ to $(-1)^x$



Multiplying $(-1)^{x+y}$



No $(-1)^{x+y}$



Homework 2

- Consider the images shown.
 - The image on the right was obtained by:
 - (a) computing the DFT and shift DC component to center
 - (b) inverse the magnitude
 - (c) computing the inverse DFT



B . I . P

- Explain (mathematically) why the image on the right appears as it does.

Homework 3

- Proving that for real image $f(x, y)$, the DFT result $F(u, v)$ satisfies property $F(u, v) = F^*(-u, -v)$
- For a 2D MR image, if the $FOV_x = FOV_y = 240mm$, and $\max(k_x) = \max(k_y) = 120 \text{ rad/m}$, Calculate the total number of full sampling pixels (total k-space sampling points) required according to Nyquist criterion?
- If we use conjugate symmetry property, calculate the minimum number of required k-space sampling pixels for problem 2.

